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(54) **3-DIMENSIONAL (3D) TISSUE-ENGINEERED
MUSCLE FOR TISSUE RESTORATION**

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(58) **Field of Classification Search**

None

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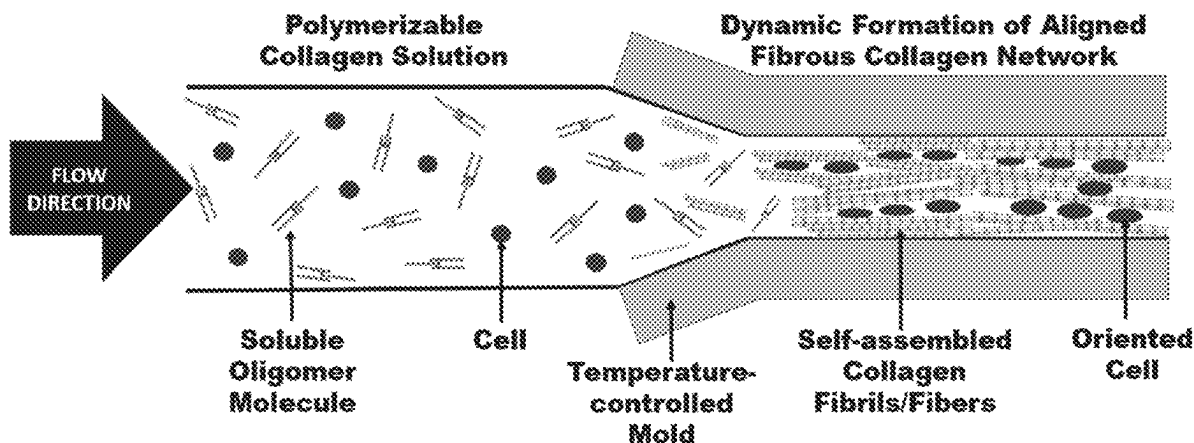
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ABSTRACT

The present disclosure provides solid collagen constructs and tissue compositions, wherein a polymerizable collagen solution or suspension is extruded in the presence or absence of cells to formed an aligned architecture comprising solid collagen constructs such as those made with fibrillar collagen. Methods of using and of manufacturing solid collagen constructs and tissue compositions, where the component collagen is solid fibrillar collagen and cells are preferentially aligned, are also provided.

20 Claims, 15 Drawing Sheets



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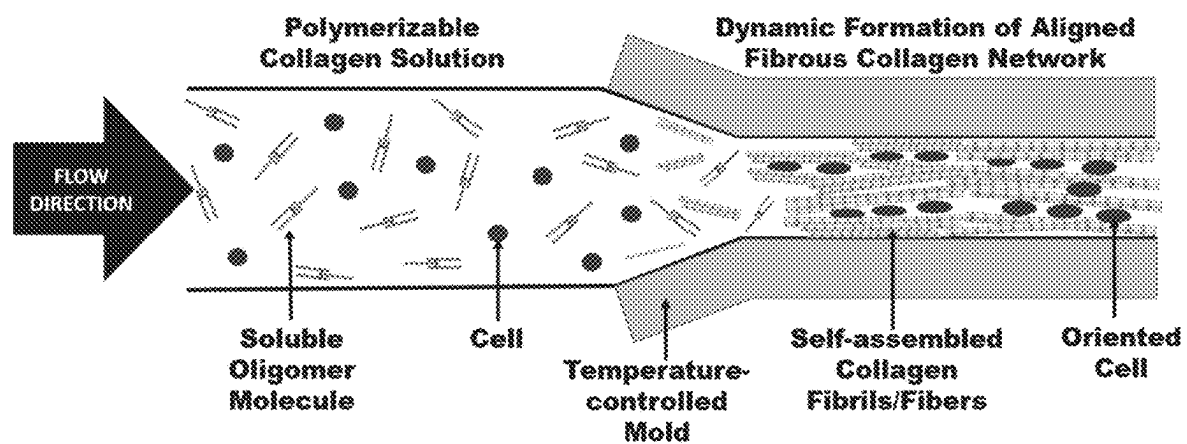


FIG. 1

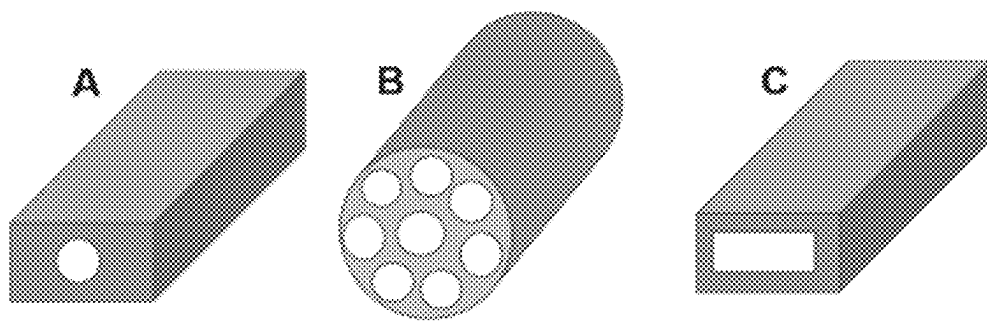


FIG. 2

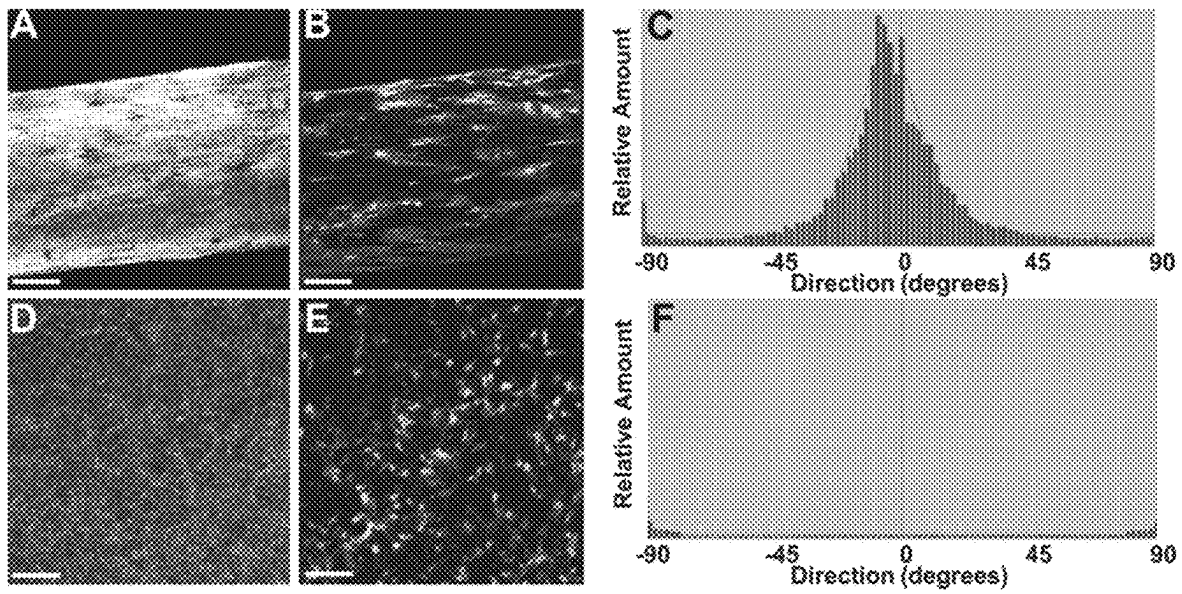


FIG. 3

FIG. 4 A

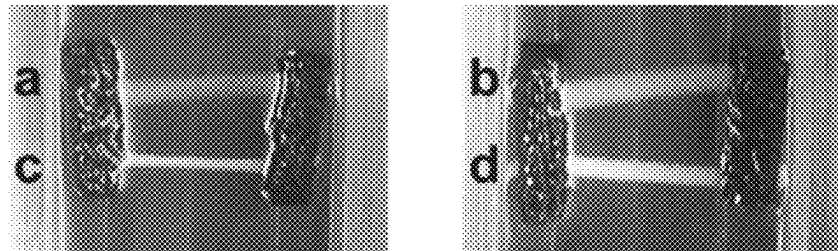
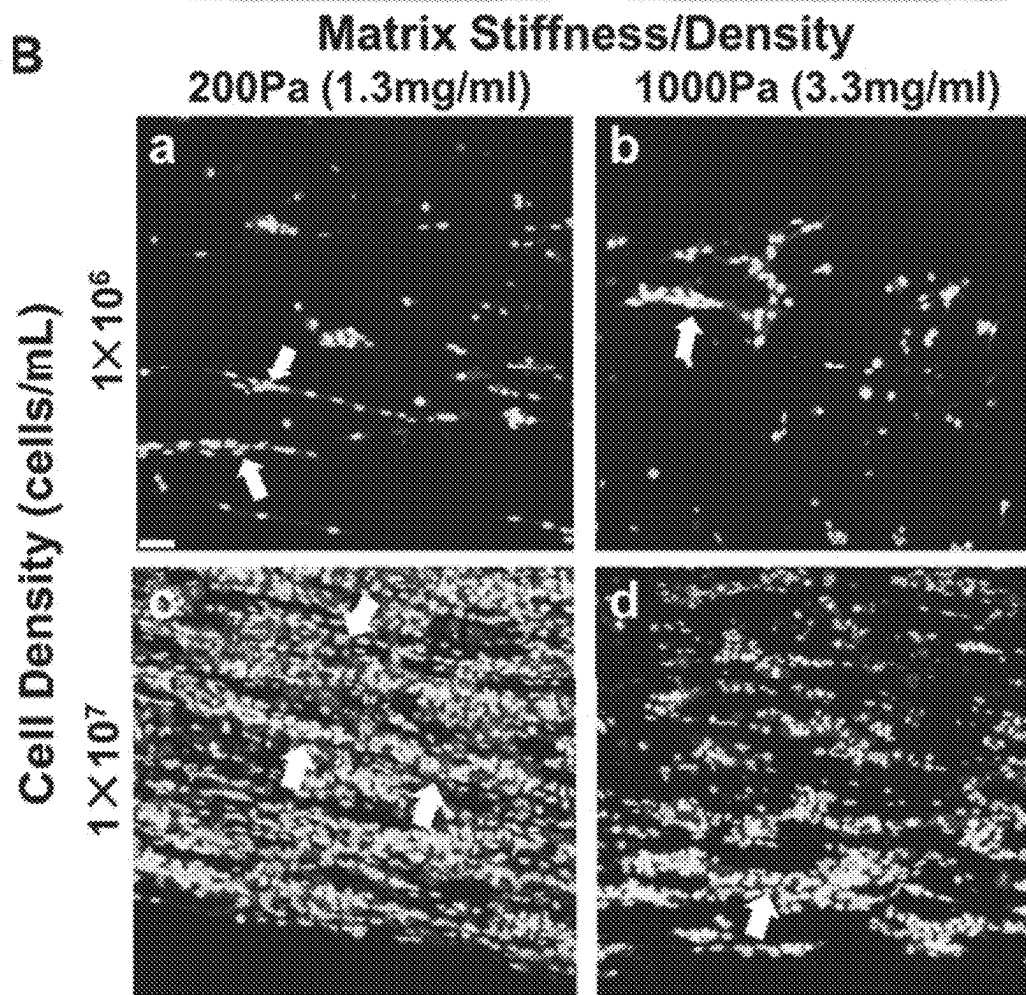


FIG. 4 B



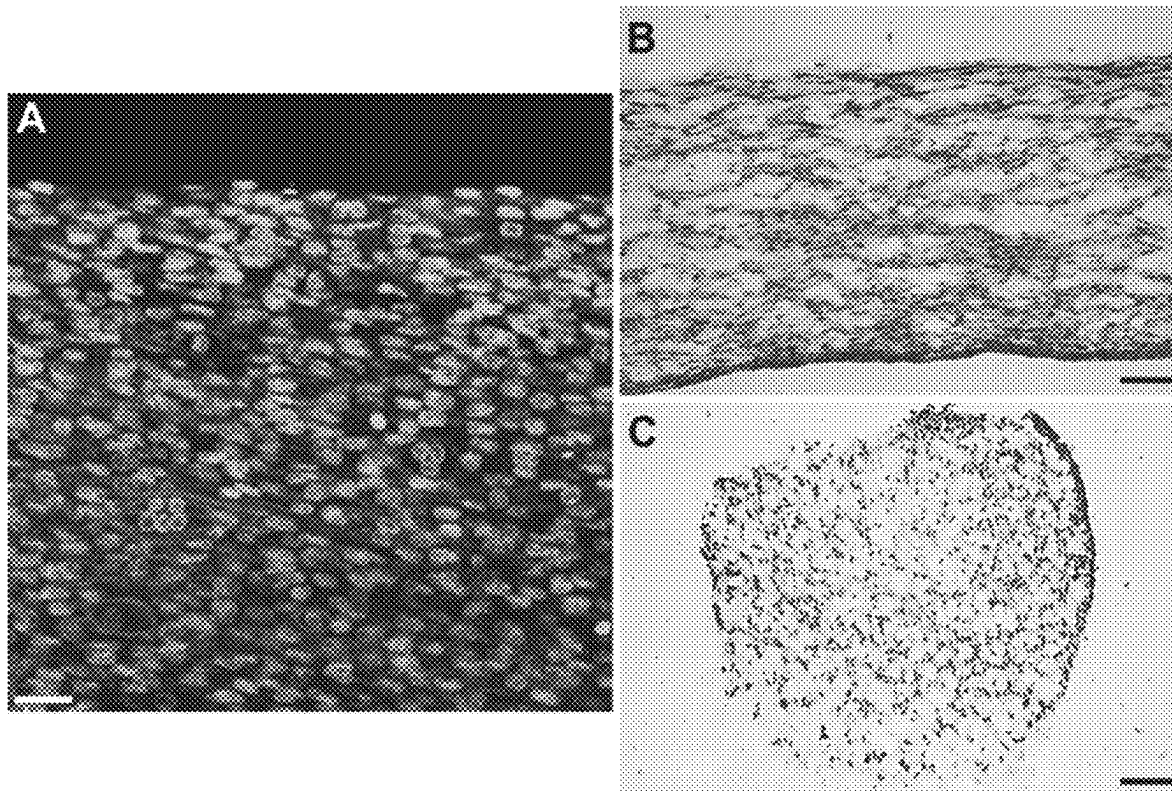


FIG. 5

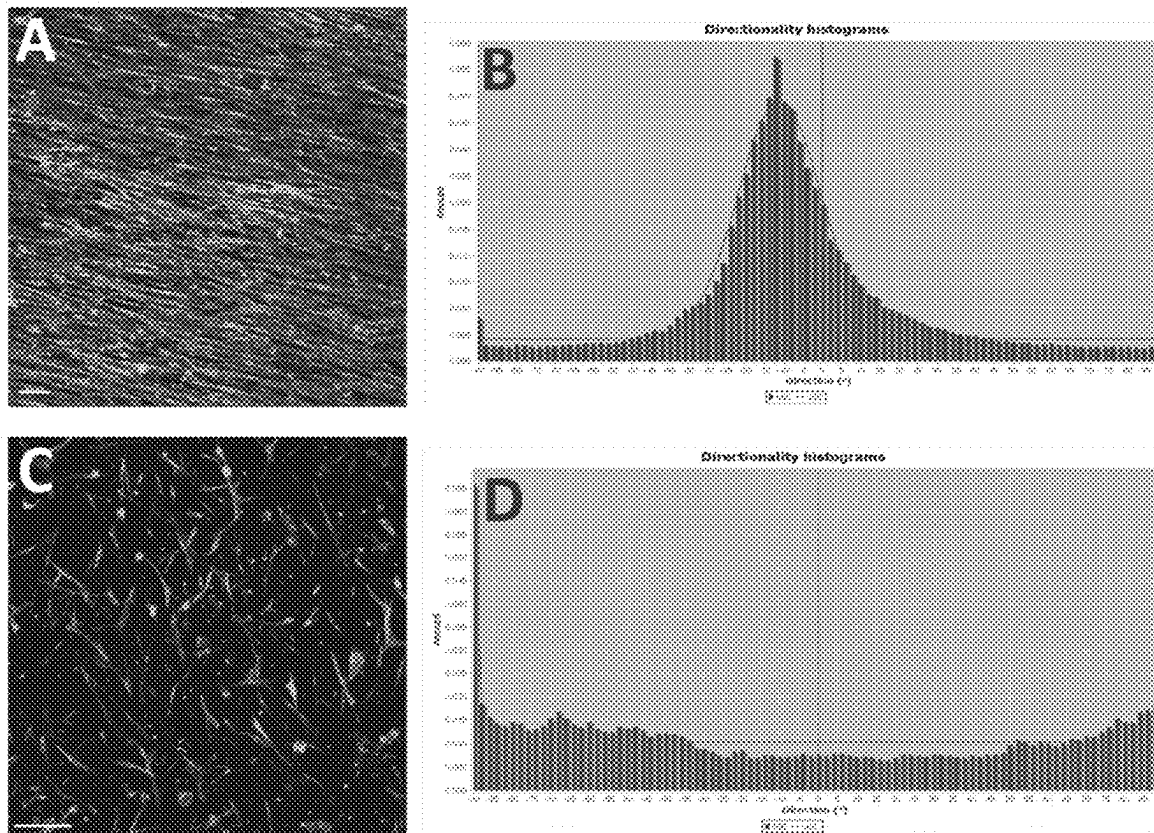


FIG. 6

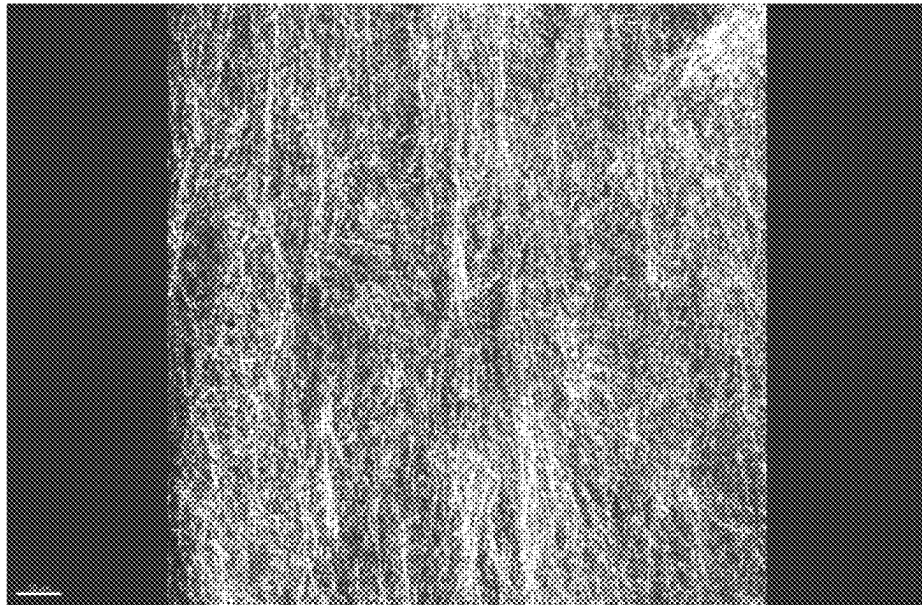


FIG. 7

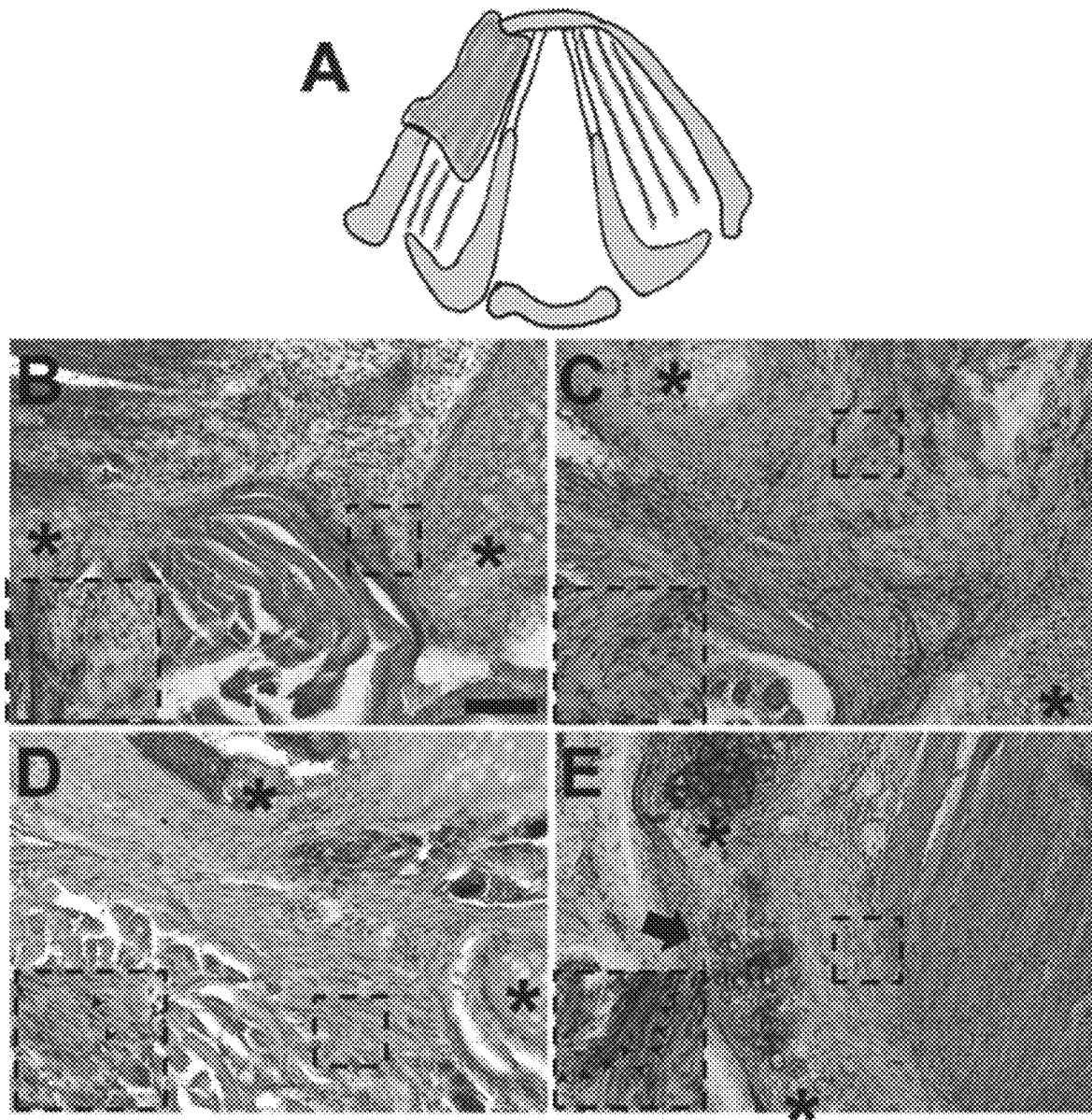


FIG. 8

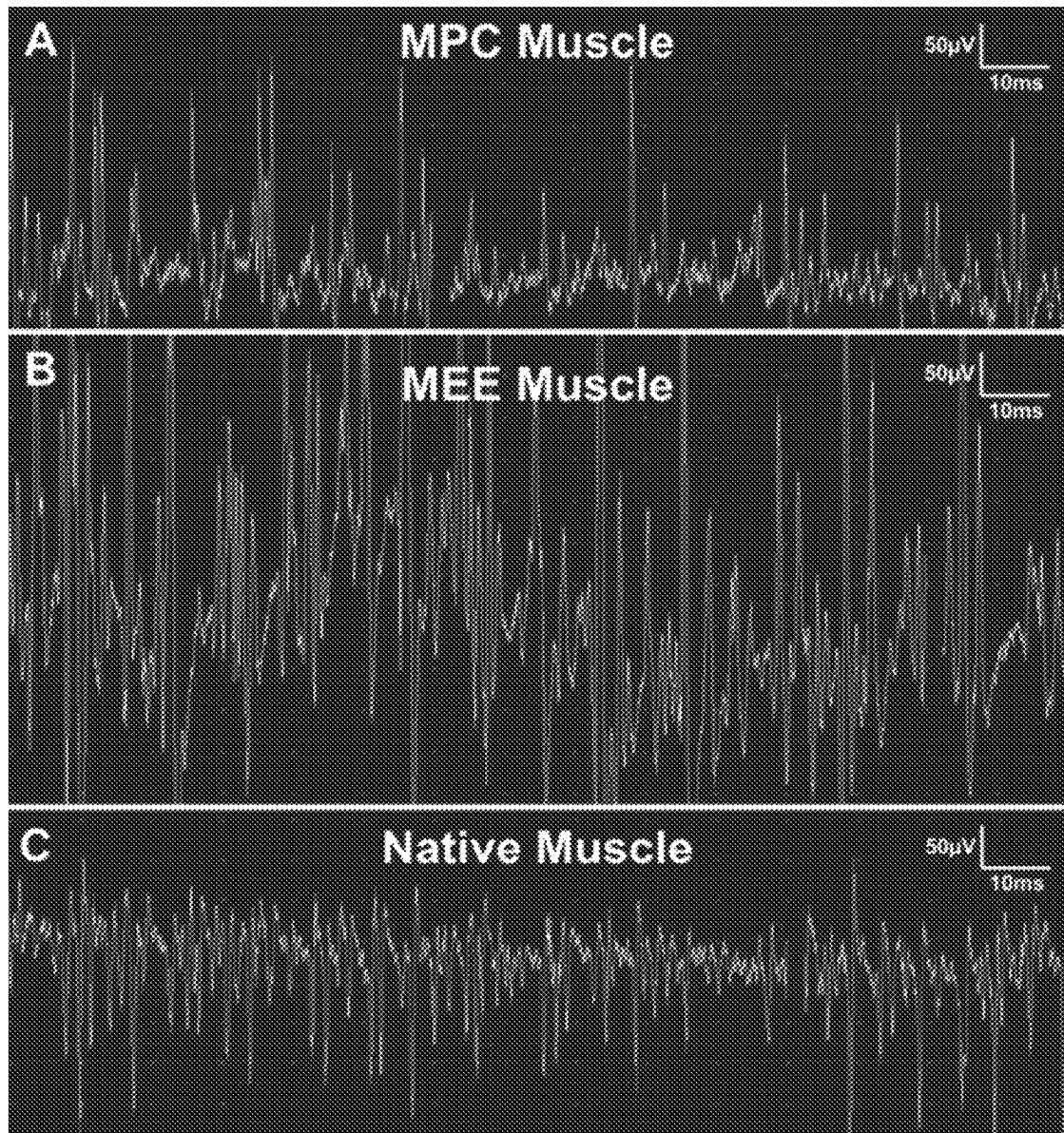


FIG. 9

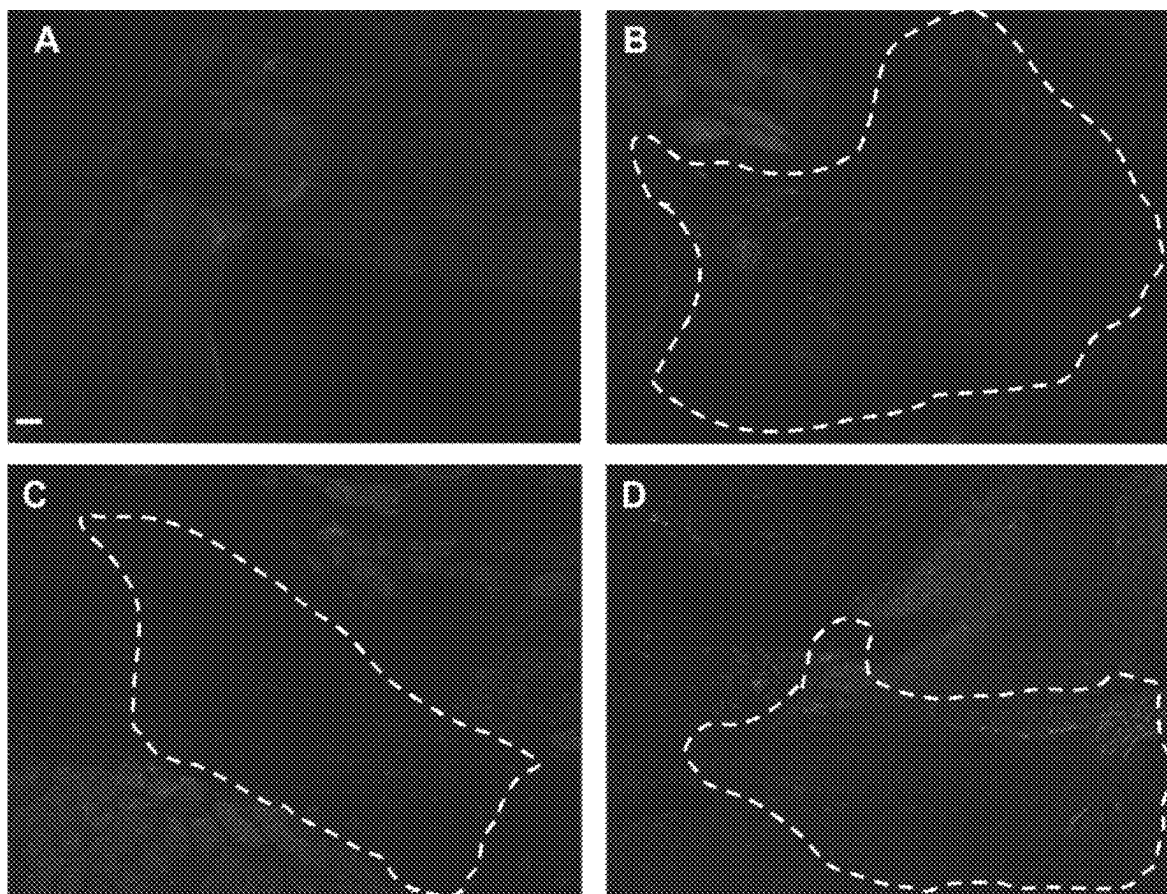


FIG. 10

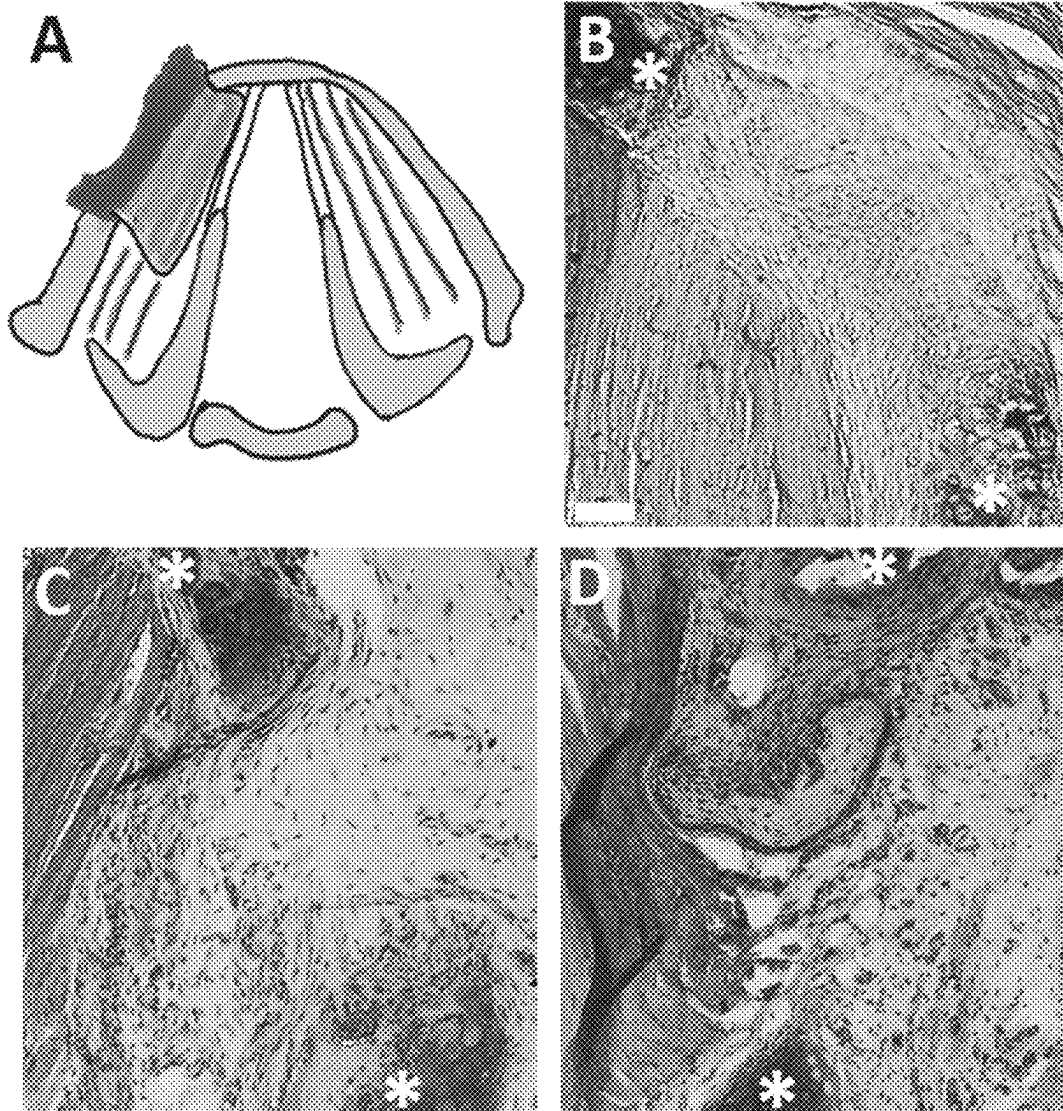


FIG. 11

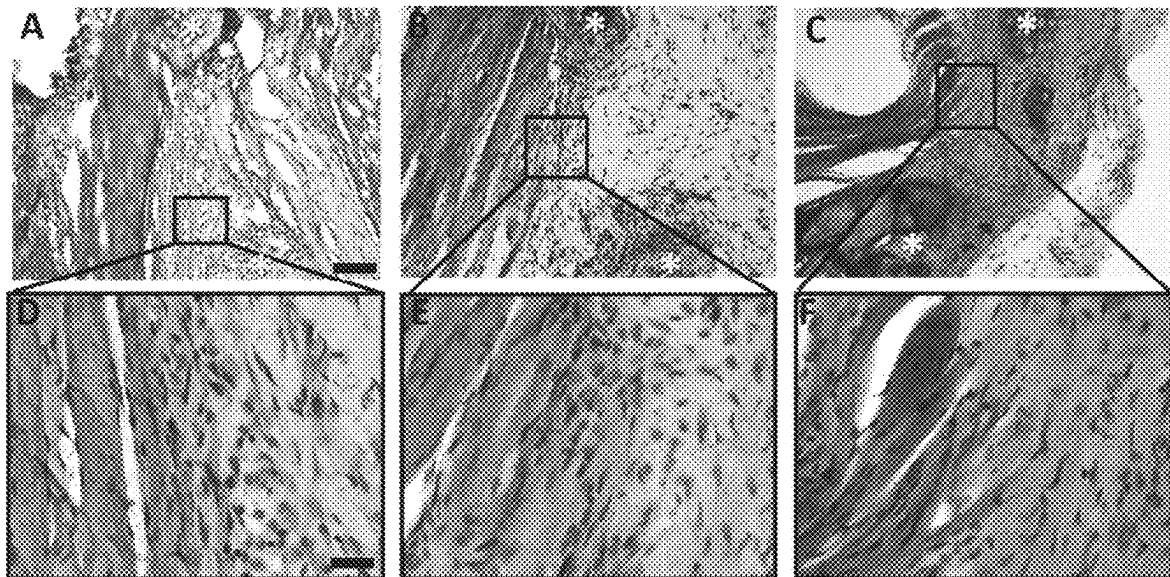


FIG. 12

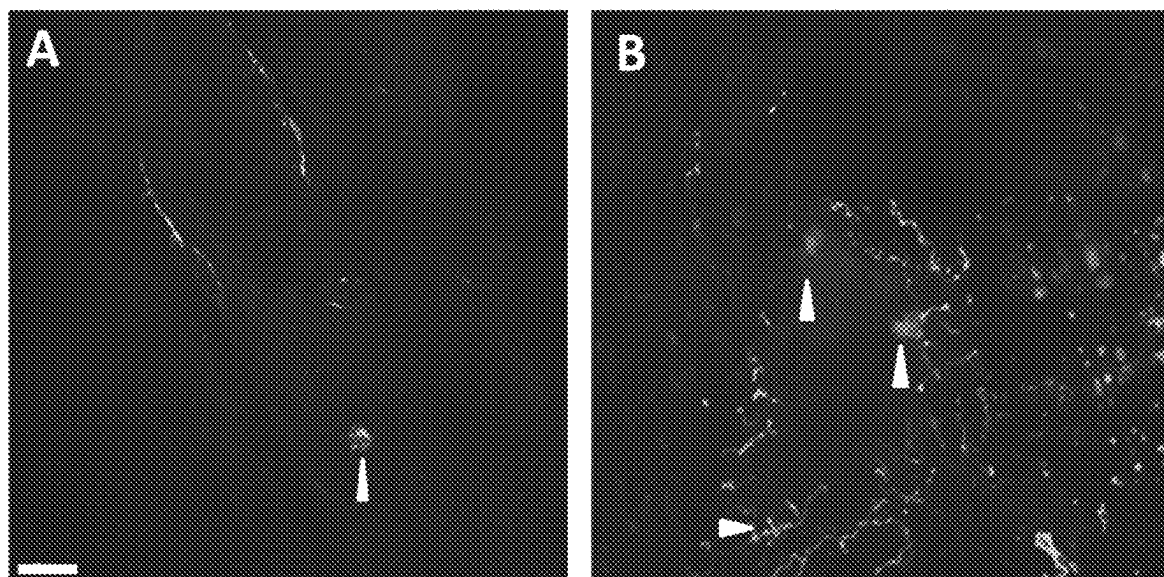


FIG. 13

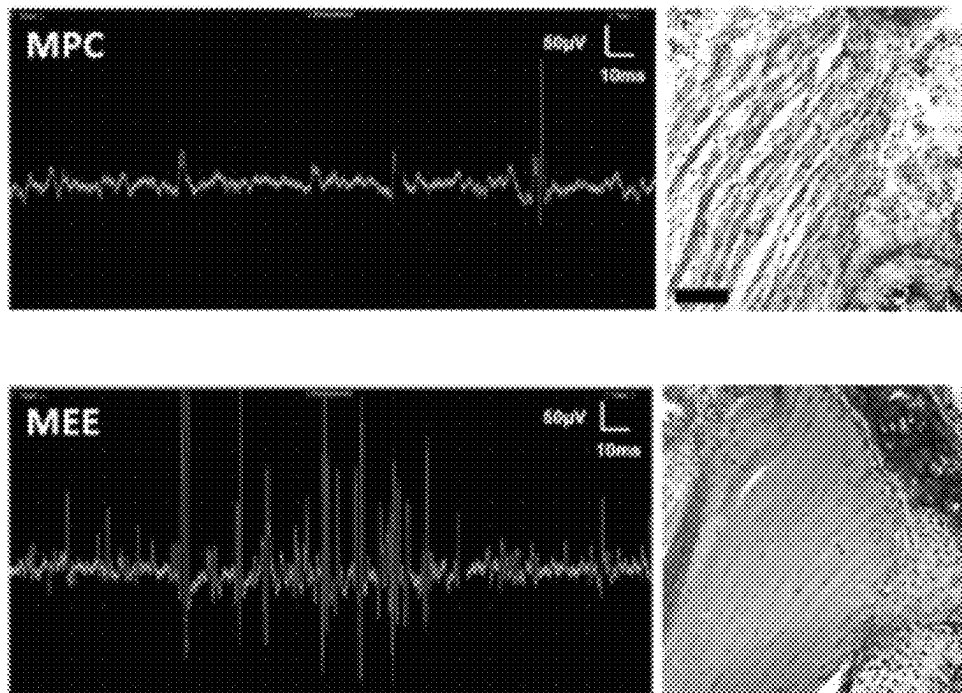


FIG. 14A

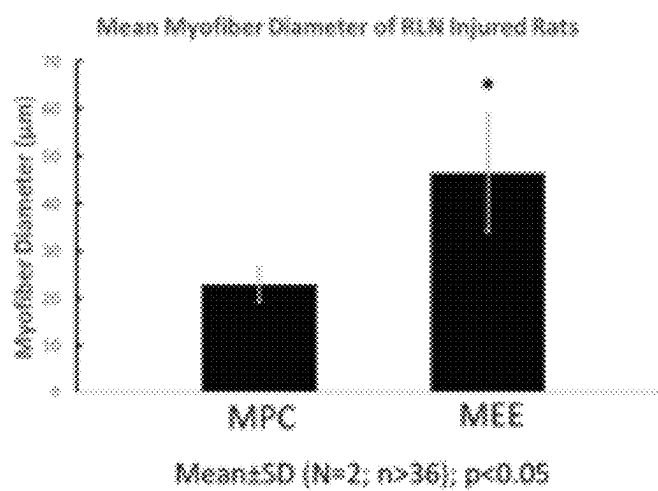


FIG. 14B

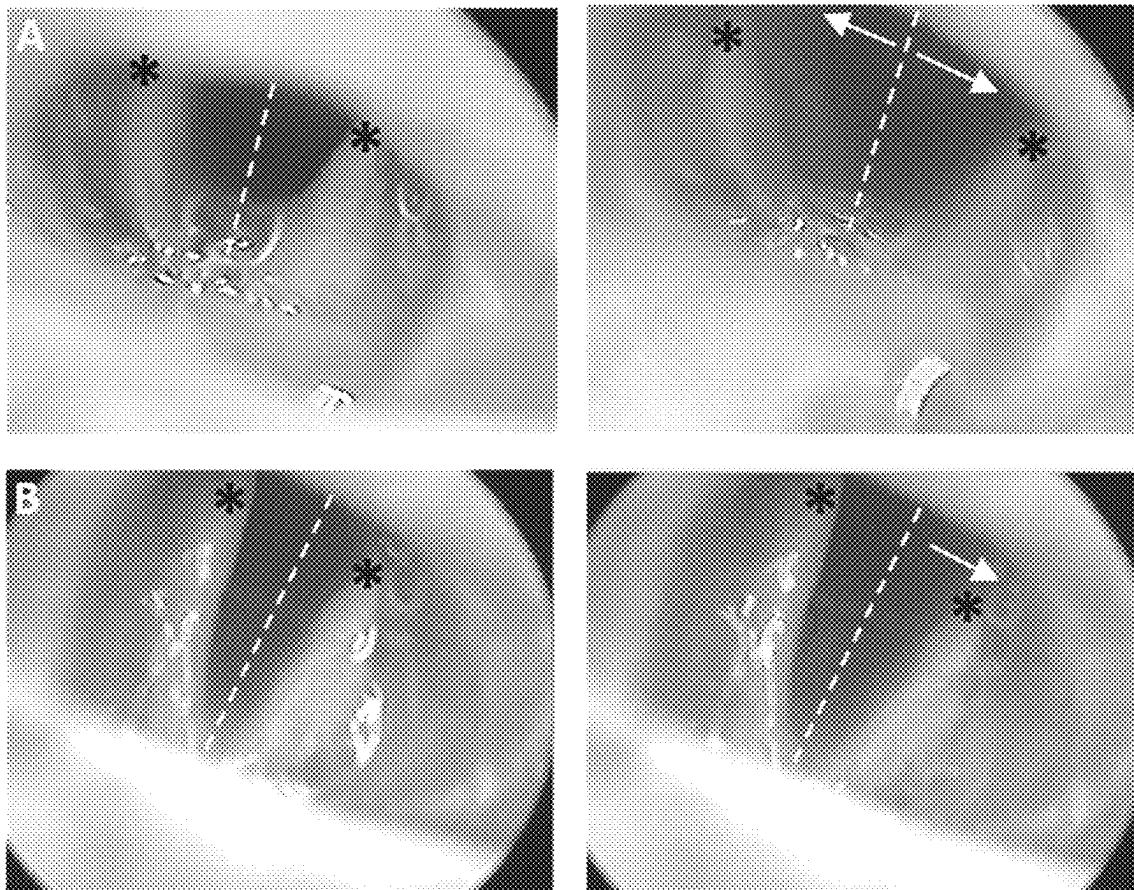


FIG. 15

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3-DIMENSIONAL (3D) TISSUE-ENGINEERED MUSCLE FOR TISSUE RESTORATION

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 16/607,665, filed on Oct. 23, 2019, which is a national stage of International Application No. PCT/US2018/029473, filed Apr. 25, 2018, which claims the benefit of and priority to U.S. Provisional Application No. 62/489,849, filed Apr. 25, 2017, the entire disclosures of which are expressly incorporated herein by reference in their entireties.

GOVERNMENT RIGHTS

This invention was made with government support under DC014070 awarded by the National Institutes of Health. The government has certain rights in the invention.

FIELD OF INVENTION

This disclosure relates to tissue restoration applications including compositions for such applications and methods for making such compositions. This disclosure includes tissue-engineered solid collagen constructs that may be made with, or without, cells used for tissue restoration as well as an in-vitro testing platform.

BACKGROUND

It has been shown that significant muscular injuries result in a reparative inflammatory-mediated healing response yielding fibrotic scar and dysfunctional muscle. For example, compromised laryngeal function, whether due to congenital malformations, trauma, cancer, or surgical defects, affects thousands of individuals worldwide each year¹. Unfortunately, therapeutic options to restore lost muscle and dynamic laryngeal functions for these patients are limited. As a result, patients suffer devastating quality of life consequences, including severe voice impairment, an ineffective or unsafe swallow, or airway obstruction, often necessitating gastrostomy and/or tracheostomy tubes. Advanced tissue engineering and regenerative strategies aimed to develop skeletal muscle implants may provide clinicians with new tools and therapeutic strategies for treating these patients.

Normal skeletal muscle reveals an intricate tissue design, including muscle fibers with their associated contractile machinery and a rich neurovascular supply which is essential for inducing and sustaining dynamic contraction. Furthermore, muscle extracellular matrix (ECM), which includes fibrillar type I collagen as a major component, plays an important role in guiding the muscle-nerve-vascular interface as well as supporting muscle's mechanical function, adaptability, and repair⁴. Therefore, it is not surprising that the majority of muscle engineering approaches focus on capturing these essential design features. At present, the most widely used cell source for engineering skeletal muscle is the putative muscle progenitor cell (MPC; satellite cell), which can be readily isolated from muscle biopsies and cultured to produce myoblasts⁵. MPC can be further modified to induce differentiation and expression of motor endplates, generating motor endplate expressing MPCs, referred to hereafter as MEE. In turn, these cells are interfaced with

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a variety of natural and synthetic biomaterials, designed to promote myoblast fusion, differentiation, and maturation in vitro or in vivo.

To achieve meaningful therapeutic benefit, it has been proposed that engineered muscle should i) be constructed from autologous cell sources, ii) recapitulate the structure and functional properties of native skeletal muscle, which represents aligned muscle fibers interfacing within an appropriate, well-organized (ECM), iii) integrate rapidly into host tissue with associated neovascularization and innervation, and iv) support scalable and patient-specific design^{2,3}.

Synthetic polymers, including polycaprolactone and poly (lactic-co-glycolic) acid, often are the engineering material of choice, largely owing to their mechanical stability, design versatility, and amenability to "additive" micro- and nano-fabrication techniques (electrospinning, patterning), where cells and materials are brought together in a stepwise fashion by adding layer upon layer. Unfortunately, upon implantation in vivo, these materials are sensed as "foreign" to cells, yielding an inflammatory-mediated, foreign-body response which is known to compromise healing and lead to poor clinical outcomes. Decellularized tissues (e.g., skeletal muscle), which are processed to maintain the complex composition, structural integrity, and architectural features of tissue extracellular matrix (ECM), also have been applied. However, these graft materials induce an inflammatory reaction as well, and their dense microstructure prevents complete recellularization and muscle recovery. Alternatively, natural polymers, such as fibrinogen, type I collagen, and Matrigel alone or in combination have been employed.

For these applications, conventional casting methods have been applied where cells are mixed with natural polymers and pipetting within molds to form cylindrical or rectangular shaped constructs with randomly organized dispersions of cells. The materials are then anchored at each end to provide passive tension, which is required to promote unidirectional cell alignment and myotube formation via cell fusion. Although it is evident that these natural polymers provide cell adhesion sites and associated bioinstructive properties, they are known to exhibit high batch-to-batch variability and are less amenable to the control of their physicochemical properties and scalable fabrication processes than synthetic polymers. Despite advancements with respect to muscle engineering, the search continues for a cost-effective and customizable muscle fabrication strategy that harnesses natural muscle formation processes (known as myogenesis) and rapid integration and neurovascular regeneration in absence of inflammation following implantation within the body.

Applications of such functional engineered muscle vary. For example, such muscle could be used to repair post-oncologic or traumatic defects, or to medialize the vocal fold in cases of paresis/paralysis. Autologous, organized, engineered muscle that has adequate bulk, integrates into host tissue, and restores tissue structure and function in absence of inflammation does not currently exist. Therefore, there is an unmet need for advanced scalable manufacturing strategies for engineered biological compositions, such as collagen and tissue compositions for tissue restoration in the presence, or in the absence of, embedded cells.

SUMMARY OF THE INVENTION

In one aspect of the disclosure, kits for tissue reconstruction comprising a polymerizable collagen solution and a

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syringe are provided. In an additional aspect of the disclosure, kits comprising a solid collagen construct and a mold are provided.

In yet an additional aspect of the disclosure, solid collagen constructs comprising aligned collagen fibrils and aligned cells are provided. In a still further aspect of the disclosure, solid collagen constructs prepared by the process of extruding a polymerizable collagen solution with an extruder to generate solid collagen constructs are provided.

In yet an additional aspect of the disclosure, solid collagen constructs prepared by the process of extruding a suspension of polymerizable collagen solution and cells with an extruder to generate solid collagen constructs are provided. In a further aspect of the disclosure, tissue implants comprising solid collagen constructs are provided.

In an additional aspect of the disclosure, processes for preparing solid collagen constructs are provided comprising extruding a polymerizable collagen solution with an extruder to generate solid collagen constructs are provided.

In still a further aspect of the disclosure, processes for preparing solid collagen constructs by the process comprising extruding a suspension of polymerizable collagen solution and cells with an extruder to generate solid collagen constructs wherein the solid collagen constructs are embedded with cells are provided.

In a further aspect of the disclosure, a 3D tissue-engineered muscle implant prepared from therapeutic cells, and type I collagen oligomers through extrusion, is provided.

Still other embodiments described in the following clause list are considered to be part of the invention.

In addition any of the embodiments described in the following clause list are considered to be part of the invention.

1. A solid collagen construct comprising aligned collagen fibrils and aligned cells.
2. A solid collagen construct prepared by the process of extruding a polymerizable collagen solution with an extruder to generate the solid collagen construct.
3. The solid collagen construct of clause 2, wherein the polymerizable collagen solution is a polymerizable collagen oligomer solution.
4. The solid collagen construct of clause 3, wherein the solution is extruded into a container.
5. The solid collagen of clause 4, wherein the container is at a higher temperature than the extruder.
6. The solid collagen construct of clause 5, wherein the extruder is at about 4° C. and the container is at about 37° C.
7. The solid collagen construct of clauses 2-6, wherein the extruder is a syringe.
8. The solid collagen construct of clauses 3-7, wherein the polymerizable collagen oligomer solution comprises collagen oligomer, water, an acid, one or more salts, and a base.
9. The solid collagen construct of clause 8, wherein the polymerizable collagen oligomer solution further comprises a sugar.
10. The solid collagen construct of clauses 8-9, wherein the acid is HCl and the base is NaOH.
11. The solid collagen construct of clauses 8-10, wherein the one or more salts are KH_2PO_4 , Na_2HPO_4 , KCl, and NaCl.
12. The solid collagen construct of clauses 9-11, wherein the sugar is glucose.

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13. The solid collagen construct of clauses 2-11, wherein the concentration of the collagen in the polymerizable collagen solution is between about 0.1 mg/ml and about 40 mg/ml.

14. The solid collagen construct of clause 13, wherein the concentration of the collagen in the polymerizable collagen solution is between about 1 mg/ml and about 10 mg/ml.

15. The solid collagen construct of clause 14, wherein the concentration of the collagen in the polymerizable collagen solution is between about 2 mg/ml and about 6 mg/ml.

16. The solid collagen construct of clause 15, wherein the concentration of the collagen in the polymerizable collagen solution is between about 3 mg/ml and about 5 mg/ml.

17. The solid collagen construct of clauses 2-16, wherein the polymerizable collagen solution is extruded at a rate of between about 1 ml/minute and about 3 ml/minute.

18. The solid collagen construct of clause 16, wherein the polymerizable collagen solution is extruded at a rate of about 2 ml/minute.

19. The solid collagen construct of clauses 2-18, wherein the pH of the polymerizable collagen solution is between about 4 and about 10.

20. The solid collagen construct of clause 19, wherein the pH of the polymerizable collagen solution is between about 6 and 8.

21. The solid collagen construct of clause 20, wherein the pH of the polymerizable collagen solution is about 7.4.

22. The solid collagen construct of clauses 2-21, wherein the solid collagen is fibrillar.

23. The solid collagen construct of clauses 2-22, wherein the polymerizable collagen solution polymerizes during extrusion in the extruder.

24. The solid collagen construct of clauses 2-23, wherein the polymerizable collagen solution polymerizes after extrusion from the extruder.

25. The solid collagen construct of clauses 4-24 wherein the container is a die or mold.

26. The solid collagen construct of clause 26, wherein the container is a mold.

27. A solid collagen construct prepared by the process comprising extruding a suspension of polymerizable collagen solution and cells with an extruder to generate a solid collagen construct wherein the solid collagen construct is embedded with cells.

28. The solid collagen construct of clause 27, wherein the polymerizable collagen solution is a polymerizable collagen oligomer solution.

29. The solid collagen construct of clauses 27-28, wherein the cells are MPC cells.

30. The solid collagen construct of clauses 27-28, wherein the cells are MEE cells.

31. The solid collagen construct of clauses 27-28, wherein the cells are ASC.

32. The solid collagen construct of clauses 27-31, wherein the polymerizable collagen oligomer suspension is extruded into a container.

33. The solid collagen construct of clause 32, wherein the container is at a higher temperature than the extruder.

34. The solid collagen construct of clause 33, wherein the extruder is at about 4° C. and the container is at about 37° C.

35. The solid collagen construct of clauses 27-34, wherein the extruder is a syringe.

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36. The solid collagen construct of clauses 28-35, wherein the polymerizable oligomer solution comprises collagen oligomer, water, an acid, one or more salts, and a base.
37. The solid collagen construct of clauses 36, wherein the polymerizable collagen oligomer solution further comprises a sugar. 5
38. The solid collagen construct of clauses 36-37, wherein the acid is HCl and the base is NaOH.
39. The solid collagen construct of clauses 36-38, wherein the one or more salts are KH_2PO_4 , Na_2HPO_4 , KCl, and NaCl. 10
40. The solid collagen construct of clauses 37-39, wherein the sugar is glucose.
41. The solid collagen construct of clauses 27-40, wherein the concentration of the collagen in the polymerizable collagen solution is between about 0.1 mg/ml and about 40 mg/ml. 15
42. The solid collagen construct of clause 41, wherein the concentration of the collagen in the polymerizable collagen solution is between about 1 mg/ml and about 6 mg/ml. 20
43. The solid collagen construct of clause 42, wherein the concentration of the collagen in the polymerizable collagen solution is between about 3 mg/ml and about 5 mg/ml. 25
44. The solid collagen construct of clauses 27-43, wherein the polymerizable collagen suspension is extruded at a rate of between about 1 ml/minute and about 3 ml/minute. 30
45. The solid collagen construct of clause 44, wherein the polymerizable collagen suspension is extruded at a rate of about 2 ml/minute.
46. The solid collagen construct of clauses 27-45, wherein the pH of the polymerizable collagen solution is between about 4 and about 10. 35
47. The solid collagen construct of clause 46, wherein the pH of the polymerizable collagen solution is between about 6 and 8.
48. The solid collagen construct of clause 47, wherein the pH of the polymerizable collagen solution is about 7.4. 40
49. The solid collagen construct of clauses 27-48, wherein the solid collagen is fibrillar.
50. The solid collagen construct of clauses 27-49, wherein the polymerizable collagen suspension polymerizes during extrusion in the extruder. 45
51. The solid collagen construct of clauses 27-50, wherein the polymerizable collagen suspension polymerizes after extrusion from the extruder.
52. The solid collagen construct of clauses 32-51 wherein the container is a die or mold. 50
53. The solid collagen construct of clause 52, wherein the container is a mold.
54. The solid collagen construct of clauses 27-53 further comprising the step of culturing the solid collagen construct. 55
55. The solid collagen construct of clauses 27-54, wherein the solid collagen construct is in the form of tissues with aligned architectures.
56. The solid collagen construct of clause 55, where the tissue is selected from cardiac muscle, nerve, smooth muscle, tendon, and ligament. 60
57. The solid collagen construct of clause 56, wherein the tissue is a muscle.
58. The solid collagen construct of clause 57, wherein the muscle is skeletal muscle, adductor muscle, cardiac muscle, or smooth muscle. 65

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59. Tissue implants for human or veterinary use comprising the solid collagen constructs of clauses 2-58.
60. Tissue implants of clause 59 wherein the tissue is selected from cardiac muscle, nerve, smooth muscle, tendon, and ligament.
61. The tissue implants of clause 60, wherein the tissue is a muscle.
62. The tissue implants of clause 61, wherein the muscle is skeletal muscle, cardiac muscle, or smooth muscle.
63. A process of preparing solid collagen constructs comprising extruding a polymerizable collagen solution with an extruder to generate a solid collagen construct.
64. The process of clause 3, wherein the polymerizable collagen solution is a polymerizable collagen oligomer solution.
65. The process of clause 64, wherein the solution is extruded into a container.
66. The solid collagen of clause 65, wherein the container is at a higher temperature than the extruder.
67. The process of clause 66, wherein the extruder is at about 4° C. and the container is at about 37° C.
68. The process of clauses 63-67, wherein the extruder is a syringe.
69. The process of clauses 64-68, wherein the polymerizable collagen oligomer solution comprises collagen oligomer, water, an acid, one or more salts, and a base.
70. The process of clause 69, wherein the polymerizable collagen oligomer solution further comprises a sugar.
71. The process of clauses 69-70, wherein the acid is HCl and the base is NaOH.
72. The process of clauses 69-71, wherein the one or more salts are KH_2PO_4 , Na_2HPO_4 , KCl, and NaCl.
73. The process of clauses 70-71, wherein the sugar is glucose.
74. The process of clauses 63-71, wherein the concentration of the collagen in the polymerizable collagen solution is between about 0.1 mg/ml and about 40 mg/ml.
75. The process of clause 74, wherein the concentration of the collagen in the polymerizable collagen solution is between about 1 mg/ml and about 10 mg/ml.
76. The process of clause 75, wherein the concentration of the collagen in the polymerizable collagen solution is between about 2 mg/ml and about 6 mg/ml.
77. The process of clause 76, wherein the concentration of the collagen in the polymerizable collagen solution is between about 3 mg/ml and about 5 mg/ml.
78. The process of clauses 63-77, wherein the polymerizable collagen solution is extruded at a rate of between about 1 ml/minute and about 3 ml/minute.
79. The process of clause 77, wherein the polymerizable collagen solution is extruded at a rate of about 2 ml/minute.
80. The process of clauses 63-79, wherein the pH of the polymerizable collagen solution is between about 4 and about 10.
81. The process of clause 80, wherein the pH of the polymerizable collagen solution is between about 6 and 8.
82. The process of clause 81, wherein the pH of the polymerizable collagen solution is about 7.4.
83. The process of clauses 63-82, wherein the solid collagen is fibrillar.
84. The process of clauses 63-83, wherein the polymerizable collagen solution polymerizes during extrusion in the extruder.

85. The process of clauses 63-84, wherein the polymerizable collagen solution polymerizes after extrusion from the extruder.
86. The process of clauses 65-85 wherein the container is a die or mold. 5
87. The process composition of clause 86, wherein the container is a mold.
88. A process for preparing solid collagen construct comprising extruding a suspension of polymerizable collagen solution and cells with an extruder to generate a solid collagen construct wherein the solid collagen construct is embedded with cells. 10
89. The process of clause 88, wherein the polymerizable collagen solution is a polymerizable collagen oligomer solution. 15
90. The process of clauses 88-89, wherein the cells are MPC cells.
91. The process of clauses 88-89, wherein the cells are MEE cells.
92. The process of clauses 88-89, wherein the cells are ASC. 20
93. The process of clauses 88-92, wherein the polymerizable collagen oligomer suspension is extruded into a container.
94. The process of clause 93, wherein the container is at a higher temperature than the extruder. 25
95. The process of clause 94, wherein the extruder is at about 4° C. and the container is at about 37° C.
96. The process of clauses 88-95, wherein the extruder is a syringe. 30
97. The process of clauses 89-96, wherein the polymerizable oligomer solution comprises collagen oligomer, water, an acid, one or more salts, and a base.
98. The process of clauses 97, wherein the polymerizable collagen oligomer solution further comprises a sugar. 35
99. The process of clauses 97-98, wherein the acid is HCl and the base is NaOH.
100. The process of clauses 97-99, wherein the one or more salts are $\text{KH}_2\text{PO}_4 \cdot \text{Na}_2\text{HPO}_4 \cdot \text{KCl}$, and NaCl.
101. The process of clauses 98-100, wherein the sugar is glucose. 40
102. The process of clauses 88-101, wherein the concentration of the collagen in the polymerizable collagen solution is between about 1 mg/ml and about 10 mg/ml and the solid collagen construct forms fibrils. 45
103. The process of clause 102, wherein the concentration of the collagen in the polymerizable collagen solution is between about 2 mg/ml and about 6 mg/ml.
104. The process of clause 103, wherein the concentration of the collagen in the polymerizable collagen solution is between about 3 mg/ml and about 5 mg/ml. 50
105. The process of clauses 88-104, wherein the polymerizable collagen suspension is extruded at a rate of between about 1 ml/minute and about 3 ml/minute.
106. The process of clause 105, wherein the polymerizable collagen suspension is extruded at a rate of about 2 ml/minute. 55
107. The process of clauses 88-106, wherein the pH of the polymerizable collagen solution is between about 4 and about 10. 60
108. The process of clause 107, wherein the pH of the polymerizable collagen solution is between about 6 and 8.
109. The process of clause 108, wherein the pH of the polymerizable collagen solution is about 7.4. 65
110. The process of clauses 88-109, wherein the solid collagen is fibrillar.

111. The process of clauses 88-110, wherein the polymerizable collagen suspension polymerizes during extrusion in the extruder.
112. The process of clauses 88-111, wherein the polymerizable collagen suspension polymerizes after extrusion from the extruder.
113. The solid collagen composition of clauses 93-112 wherein the container is a die or mold.
114. The solid collagen composition of clause 113, wherein the container is a mold.
115. The process of clauses 88-114, further comprising the step of culturing the solid collagen construct.
116. The process of clauses 88-115, wherein the solid collagen construct is in the form of tissues with aligned architectures.
117. The process of clause 116, where the tissue is selected from cardiac muscle, nerve, smooth muscle, tendon, and ligament.
118. The process of clause 117, wherein the tissue is a muscle.
119. The process of clause 118, wherein the muscle is skeletal muscle, cardiac muscle, or smooth muscle.
120. The solid collagen compositions of clause 41, wherein the concentration of collagen in the polymerizable collagen solution is between about 1 mg/ml and about 10 mg/ml.
121. The solid collagen composition of clauses 24 and 51, wherein polymerization occurs after extrusion from the extruder.
122. The process of clauses 85 and 112 wherein polymerization occurs after extrusion from the extruder.
123. A kit for tissue reconstruction comprising a polymerizable collagen solution or a polymerizable collagen suspension and a syringe.
124. The kit of clause 123, wherein the polymerizable collagen solution is a solution of polymerizable collagen oligomers.
125. A kit for tissue reconstruction comprising a solid collagen construct and a mold.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Schematic showing extrusion of polymerizable type I collagen oligomer suspension containing cells through a temperature-controlled mold for rapid production of engineered tissue constructs. During extrusion the soluble oligomer molecules undergo thermally controlled polymerization to form oriented solid fibrils which further associate to form aligned composites comprised of solid fibrillar collagen and cells. Extrusion can also be done in the absence of cells.

FIG. 2: Examples of extrusion mold geometries used to create engineered collagen and tissue constructs.

FIG. 3: Representative images and directionality analyses for engineered muscle constructs formed using extrusion method (A-C) compared to conventional casting technique (D-F). Fabricated muscle constructs were cultured 24 hours and analyzed by confocal reflection microscopy for visualization of fibril microstructure (A,D), confocal fluorescence microscopy for visualization of myoblast morphology (B,E), and ImageJ Directionality algorithm for alignment determination (C,F). Scale bar=50 μm

FIG. 4A. Engineered muscle construct culture and optimization. Muscle constructs were formed by extrusion of C2C12 myoblasts at 10^6 cells/mL (a,b) or 10^7 cells/mL (c,d) suspended within type I collagen oligomer solutions prepared at 1.3 mg/mL (ac) and 3.3 mg/mL (b,d). A.

Constructs were cultured within a custom culture device for 14 days under passive tension.

FIG. 4B. Confocal images (100 μ m thickness) of 14-day constructs stained with phalloidin (F-actin) and Draq5 (nucleus) show more extensive myotube formation (white arrows) for constructs prepared at high cell densities (10E7 cells/mL) and low oligomer concentrations (1.3 mg/mL). Scale bar=50 μ m

FIG. 5. Engineered muscle constructs (14 days) formed by extrusion of F344 rat primary MPCs (10E7 cells/mL) suspended within type I collagen oligomer solutions prepared at 1.3 mg/mL. A. Confocal images (50 μ m thickness) of 14-day constructs stained with phalloidin (F-actin) and Draq5 (nucleus) show highly aligned cell morphology and fused myotube formation. Cells were distributed throughout the entire construct volume as indicated by H&E staining of longitudinal (B) and transverse (C) sections. Scale bars=30 μ m (A), 500 μ m (B-C)

FIG. 6: Representative images and directionality analyses for engineered tissue constructs formed with adipose-derived stem cells (ASCs) suspended in type I collagen oligomer solution using extrusion method (A-B) compared to conventional casting technique (C-D). Fabricated constructs were cultured 24 hours, stained with phalloidin (F-actin) and Draq5 (nucleus) and analyzed by confocal fluorescence microscopy for visualization of cell morphology (A,C) and ImageJ Directionality algorithm for alignment determination (B,D). Scale bars=100 μ m.

FIG. 7: Representative confocal reflection microscopy image of engineered collagen construct with aligned collagen fibrils formed by extrusion of a type I collagen oligomer solution prepared at a concentration of 3.3 mg/mL. Scale bar=10 μ m.

FIG. 8. (A) Schematic showing partial laryngectomy model, which involved removal of a section of hemi-thyroid cartilage and associated adductor muscle from left side. Cross-section of rat larynx showing surgical procedure and implant placement. (B-E) Representative H&E stained sections of untreated control defect (B, 2 months) and defects treated with oligomer-only implant (C, 2 months), engineered muscle implant (D, 1 month), and engineered muscle implant (E, 3 months). Untreated defect (B) showed gap filled with native sternohyoid muscle and healing via inflammation and fibrotic scar tissue formation. Oligomer-only group showed no significant inflammatory response and construct populated with mesenchymal cells. Engineered muscle implants showed progressive increase in volume of striated muscle (boxed areas and inserts; inserts represent high magnification of boxed areas) and cartilage regeneration (arrow) over time with similar responses observed for both MPC- and MEE-oligomer implants. Asterix (*) indicate edges of cartilage defect. Scale bar=200 μ m

FIG. 9. Representative electromyography (EMG) tracings captured at 50 μ V amplitude and 10 ms sweep speeds during active laryngospasm to detect firing within engineered muscle implant 3 months following implantation (A,B) or native adductor muscle complex (C). A. Defect treated with engineered muscle prepared with MPCs (MPC Muscle) generated abundant, variable-sized motor unit potentials. B. Engineered muscle prepared with motor endplate expressing MPCs (MEE Muscle) generated potentials that were significantly larger in amplitude and frequency compared to MPC Muscle and native adductor muscle. C. Native adductor muscle complex demonstrated bursts of motor unit potentials during laryngospasm.

FIG. 10. Anti-beta III tubulin and α -bungarotoxin staining of (A) untreated control defect, (B) oligomer-only control,

(C) MSC-oligomer construct, and (D) MEE-oligomer construct at 3 months following surgical implantation post-op. Dotted line outlines approximate construct boundaries. The MEE-oligomer construct (D) showed increased innervation when compared to the MEE-oligomer construct (C). Both the MEE- and MSC-oligomer constructs exhibited increased innervation when compared to the oligomer-only and untreated controls. Scale bar=100 μ m

FIG. 11: (A) Schematic showing partial laryngectomy model, which involved removal of a section of hemi-thyroid cartilage and associated underlying adductor muscle in the presence or absence of recurrent laryngeal nerve injury. Cross-section of rat larynx showing surgical reconstruction with tissue-engineered cartilage and muscle implants. The cartilage constructs showed progressive remodeling, increased proteoglycan deposition, and maturation over the 1, 3, and 6 month timepoints (B, C, D respectively). Asterix represents edges of surgically created defect. Scale Bar=100 μ m

FIG. 12: H&E staining shows progressive restoration of partial laryngectomy defect reconstructed with engineered muscle constructs at 1 month (A,D), 3 months (B,E), and 6 months (C,F), marked by muscular integration, regeneration, and maturation with no inflammatory response. Asterix marks edges of surgical defect. Scale Bar: A-C 100 μ m; D-F 25 μ m

FIG. 13: α -bungarotoxin and beta III tubulin staining shows presence of motor endplates and significant innervation of the engineered muscle constructs after 6 months. A: MSC-oligomer constructs showed a lesser degree of innervation B. MEE-oligomer constructs showed robust innervation and motor endplate expression (arrowheads). Scale Bar=10 μ m

FIG. 14A: Animals with injured recurrent laryngeal nerves (RLN) treated with MEE-oligomer constructs showed increased EMG activity and myofiber diameter compared to MPC-oligomer implants. EMG of the adductor muscle complex showed increased activity in MEE-oligomer (C) treated animals compared to MPC-oligomer (A) treated animals. Histology showed some muscle atrophy in MPC-oligomer (B) treated animals and near normal myofiber architecture in MEE-oligomer (D) treated animals. Myofiber diameter of adductor muscle (E) showed significant increase in MEE-oligomer treated animals (Mean $\pm$ SD, p<0.001, 2 animals per group, minimum 36 measurements per animal). Scale bar=100 μ m.

FIG. 14B: Bar graph illustrating mean myofiber diameter of adductor muscle following hemilaryngeal reconstruction within RLN injured rats with MPC-oligomer constructs or MEE-oligomer constructs.

FIG. 15: Still images from video laryngoscopy obtained 3 months after hemilaryngeal reconstruction in rat laryngectomy model with recurrent laryngeal nerve injury. Recovery of left vocal fold movement was observed in animals treated with MEE-oligomer constructs (A). Animals treated with the MSC-oligomer constructs (B) showed less pronounced movement in the left vocal fold.

DETAILED DESCRIPTION

While the concepts of the present disclosure are illustrated and described in detail in the figures and the description herein, results in the figures and their description are to be considered as exemplary and not restrictive in character; it being understood that only the illustrative embodiments are

shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Unless defined otherwise, the scientific and technology nomenclatures have the same meaning as commonly understood by a person in the ordinary skill in the art pertaining to this disclosure.

In many embodiments of the disclosure, solid collagen constructs, and processes for making them, are provided wherein the solid collagen constructs are prepared by extruding a polymerizable collagen solution with an extruder thereby generating a solid collagen construct. The polymerizable collagen solution is often a solution wherein the collagen contains oligomers such that the polymerizable collagen solution is an polymerizable solution of oligomeric collagen. In many embodiments, the collagen is only or primarily oligomeric collagen and thus contains no or substantially no monomeric collagen. The extrusion is often done into a container such as a dye or mold. Often the extrusion is done in an extruder, such as a syringe, which is kept on ice (e.g., at about 4° C.). Although the polymerizable solution may polymerize at such temperatures, the polymerization conditions are kept such that extrusion is still possible and practical. When extruded into a container such as a mold or die, the container temperature is often kept at elevated temperatures whether room temperature or body temperature (e.g., 37° C.) to accelerate polymerization.

The polymerizable solution includes or contains components such that polymerization can be initiated at 4 C and accelerated at higher temperatures. For example, a typical polymerizable collagen oligomer solution may be prepared in accordance with Example 1. In many embodiments, the polymerizable collagen oligomer solution contains collagen oligomers, such as a type 1 collagen oligomer, which has been dissolved in water and acid, such as HCl.

The pH is raised in the presence of one or more salts and base, such as NaOH. A sugar may optionally be added, such as glucose. An example of a combination of salts which is the one or more salts is $\text{KH}_2\text{PO}_4 \cdot \text{Na}_2\text{HPO}_4 \cdot \text{KCl}$, and NaCl. For example, when a solubilized collagen oligomeric solution is acidic and is then neutralized by the one or more salts and the base such that the pH increases to the range of between about 4 and 10 including within about 6 and about 8, and further including about 7.4, polymerization spontaneously occurs with the rate of such polymerization being dependent upon temperature.

Unless explicitly defined otherwise, the term, “solid collagen construct,” refers to collagen compositions made, for example, by extrusion, in accordance with the disclosure.

Some materials that may be used to practice some embodiments of the invention can be found in U.S. Pat. No. 9,878,071 B2 issued on Jan. 30, 2018 and incorporated fully herein by reference.

In some embodiments, the polymerizable collagen composition, which may be a solution or a suspension, for example may be prepared under various conditions. For example, factors such as pH, phosphate concentration, temperature, buffer composition, ionic strength, and composition and concentration of the extracellular matrix components which includes both collagen and non-collagenous molecules, may be varied by additives or changing the environmental conditions of the polymerizable collagen composition. Examples of additives include nutrients, including minerals, amino acids, sugars, peptides, proteins, vitamins (such as ascorbic acid), or glycoproteins that facilitate hematopoietic stem cell culture, such as laminin and fibronectin, hyaluronic acid, or growth factors such as

platelet-derived growth factor, or transforming growth factor beta, and glucocorticoids such as dexamethasone. Other additives include fibrillogenesis inhibitors, such as glycerol, glucose, or polyhydroxylated compounds can be added prior to or during polymerization. Additional additives include cross-linking agents, such as carbodiimides, aldehydes, lysloxidase, N-hydroxysuccinimide esters, imidoesters, hydrazides, and maleimides, and the like can be added before, during, or after polymerization.

With regards to sourcing the collagen starting material, it may be solubilized from tissue. For example, the collagen can be prepared by utilizing acid-solubilized collagen and defined polymerization conditions that are controlled to yield three-dimensional collagen matrices with a range of controlled assembly kinetics (e.g., polymerization half-time), molecular compositions, and fibril microstructure-mechanical properties, for example, as described in U.S. patent application Ser. No. 11/435,635 (published Nov. 22, 2007, as Publication No. 2007-0269476 A1) and Ser. No. 11/903,326 (published Oct. 30, 2008, as Publication No. 2008-0268052), each incorporated herein by reference in its entirety. In other embodiments, the collagen is polymerizable collagen. In yet other embodiment, the collagen is Type I collagen.

In some embodiments, the sourced collagen starting material is unnatural collagen. As used herein, the phrase “unnatural collagen” refers to collagen that has been removed from a source tissue. Optionally, the unnatural collagen may be solubilized from the tissue source. In other embodiments, the collagen is synthetic collagen. In yet other embodiments, the collagen is recombinant collagen.

In one aspect, unnatural collagen or collagen components can be used and can be obtained from a number of sources, including for example, porcine skin, to construct the collagen compositions described herein. Suitable tissues useful as a collagen-containing source material for isolating collagen or collagen components to make the collagen compositions described herein are submucosa tissues or any other extracellular matrix-containing tissues of a warm-blooded vertebrate. Suitable methods of preparing submucosa tissues are described in U.S. Pat. Nos. 4,902,508; 5,281,422; and 5,275,826, each incorporated herein by reference. Extracellular matrix material-containing tissues other than submucosa tissue may be used to obtain collagen in accordance with the methods and compositions described herein. Methods of preparing other extracellular matrix material-derived tissues for use in obtaining purified collagen or partially purified extracellular matrix components are known to those skilled in the art. For example, see U.S. Pat. No. 5,163,955 (pericardial tissue); U.S. Pat. No. 5,554,389 (urinary bladder submucosa tissue); U.S. Pat. No. 6,099,567 (stomach submucosa tissue); U.S. Pat. No. 6,576,265 (extracellular matrix tissues generally); U.S. Pat. No. 6,793,939 (liver basement membrane tissues); and U.S. patent application publication no. US-2005-0019419-A1 (liver basement membrane tissues); and international publication no. WO 2001/45765 (extracellular matrix tissues generally), each incorporated herein by reference. In various other embodiments, the collagen-containing source material can be selected from the group consisting of placental tissue, ovarian tissue, uterine tissue, animal tail tissue, and skin tissue. In some embodiments, the collagen is selected from the group consisting of pig skin collagen, bovine collagen, and human collagen. Any suitable extracellular matrix-containing tissue can be used as a collagen-containing source material to isolate purified collagen or partially purified extracellular matrix components.

An illustrative preparation method for preparing submucosa tissues as a source of purified collagen or partially purified extracellular matrix components is described in U.S. Pat. No. 4,902,508, the disclosure of which is incorporated herein by reference. In one embodiment, a segment of vertebrate intestine, for example, preferably harvested from porcine, ovine or bovine species, but not excluding other species, is subjected to abrasion using a longitudinal wiping motion to remove cells or cell-removal is accomplished by hypotonic or hypertonic lysis. In one embodiment, the submucosa tissue is rinsed under hypotonic conditions, such as with water or with saline under hypotonic conditions and is optionally sterilized. In another illustrative embodiment, such compositions can be prepared by mechanically removing the luminal portion of the tunica mucosa and the external muscle layers and/or lysing resident cells with hypotonic or hypertonic washes, such as with water or saline. In these embodiments, the submucosa tissue can be stored in a hydrated or dehydrated state prior to isolation of the purified collagen or partially purified extracellular matrix components. In various aspects, the submucosa tissue can comprise any delamination embodiment, including the tunica submucosa delaminated from both the tunica muscularis and at least the luminal portion of the tunica mucosa of a warm-blooded vertebrate.

In some embodiments, the collagen is oligomeric collagen. The presence of oligomeric collagen enhances the self-assembly potential by increasing the assembly rate and by yielding collagen compositions with distinct fibril microstructures and increased mechanical integrity (e.g., stiffness). In some embodiments, the collagen comprises oligomeric collagen. In other embodiments, the collagen consists essentially of oligomeric collagen. In yet other embodiments, the collagen consists of oligomeric collagen.

In some embodiments, the collagen is monomeric collagen. In some embodiments, the collagen is atelocollagen. As used herein, the term "atelocollagen" refers to collagen that is treated in vitro with pepsin or another suitable protease or agent to eliminate or substantially reduce telopeptide regions which contain intermolecular cross-linking sites. In other embodiments, the monomeric collagen is telocollagen. As used herein, the term "telocollagen" refers to acid solubilized collagen that retains its telopeptide ends.

In certain embodiments, the collagen comprises oligomeric collagen and atelocollagen. In other embodiments, the collagen comprises oligomeric collagen, monomeric collagen, and atelocollagen. The amounts of oligomeric collagen, monomeric collagen, and/or atelocollagen may be formulated in the collagen compositions to advantageously maximize the stiffness, strength, fluid and mass transport, proteolytic degradation or compatibility of the engineered collagen compositions.

In any of the embodiments described herein, the collagen can have a predetermined percentage of collagen oligomers. In various embodiments, the predetermined percentage of collagen oligomers can be about 0.5% to about 100%, about 30% to about 100%, about 40% to about 100%, about 50% to about 100%, about 60% to about 100%, about 70% to about 100%, about 80% to about 100%, about 90% to about 100%, about 95% to about 100%, or about 100%. In yet another embodiment, the collagen oligomers are obtained from a collagen-containing source material enriched with collagen oligomers (e.g., pig skin).

In any of the embodiments described herein, the collagen sourced as starting material can have an oligomer content quantified by average polymer molecular weight (AMW). As described herein, modulation of AMW can affect polym-

erization kinetics, fibril microstructure, molecular properties, and fibril architecture of the matrices, for example, interfibril branching, pore size, and mechanical integrity (e.g., matrix stiffness). In another embodiment, the oligomer content of the purified collagen, as quantified by average polymer molecular weight, positively correlates with matrix stiffness.

In some embodiments, the collagen is reduced collagen. As used herein "reduced collagen" means collagen that is reduced in vitro to eliminate or substantially reduce reactive aldehydes. For example, collagen may be reduced in vitro by treatment of collagen with a reducing agent (e.g., sodium borohydride).

In some embodiments, the collagen is oligomer 260 collagen. As used herein "oligomer 260 collagen" is a collagen preparation made (e.g., from porcine skin), by procedures resulting in isolation of oligomers, where the collagen preparation has a prominent band at molecular weight 260, where the band is not prominent or is lacking in corresponding monomer preparations. The presence of the band can be determined by SDS polyacrylamide gel electrophoresis. Oligomer 260 collagen is further described U.S. patent application Ser. No. 13/192,276 (published Feb. 2, 2012, as Publication No. 2012-0027732 A1), incorporated herein by reference.

The solid collagen constructs herein described may be made under controlled conditions, such as by extrusion, and by changing extrusion parameters such as, for example, extrusion rate, geometry of the container (e.g., mold), viscosity of polymerizable solution or suspension, presence or absence of additives, cell density, temperature, porosity of extrusion container (e.g., mold) to obtain particular physical properties. For example, the solid collagen constructs may have desired collagen fibril density, pore size (fibril-fibril branching), elastic modulus, tensile strain, tensile stress, linear modulus, compressive modulus, loss modulus, fibril area fraction, fibril volume fraction, collagen concentration, cell seeding density, shear storage modulus (G' or elastic (solid-like) behavior), and phase angle delta (δ , or the measure of the fluid (viscous)—to solid (elastic)-like behavior; δ equals 0.degree. for Hookean solid and 90.degree. for Newtonian fluid).

As used herein, a "modulus" can be an elastic or linear modulus (defined by the slope of the linear region of the stress-strain curve obtained using conventional mechanical testing protocols; i.e., stiffness), a compressive modulus, a loss modulus, or a shear storage modulus (e.g., a storage modulus). These terms are well-known to those skilled in the art.

As used herein, a "fibril volume fraction" (i.e., fibril density) is defined as the percent area of the total area occupied by fibrils in three dimensions.

In any embodiment described herein, the fibril volume fraction of the collagen composition is about 1% to about 60%. In various embodiments, the collagen composition can contain fibrils with specific characteristics, for example, a fibril volume fraction (i.e., density) of about 2% to about 60%, about 2% to about 40%, about 5% to about 60%, about 15% to about 60%, about 2% to about 30%, about 5% to about 30%, about 15% to about 30%, or about 20% to about 30%.

It may be desirable to control or identify the concentration of the collagen, such as the collagen oligomer in solution, since different concentrations may yield solid collagen constructs with different properties. Typical ranges of collagen concentrations in the polymerizable collagen solutions, such as polymerizable oligomer collagen solutions, include

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between about 0.1 mg/ml and about 40 mg/ml, including between about 1 mg/ml and about 10 mg/ml, including between about 2 mg/ml and about 6 mg/ml, including between about 3 mg/ml and about 5 mg/ml. The rate of extrusion through the extruder such as, for example, a syringe, may also be controlled. Exemplary rates include between about 1 ml/min and about 3 ml/min including, or example, about 2 ml/min.

When the solid construct polymerizes, whether in the syringe during extrusion, or after exiting the extruder such as in a container (such as a die or mold), or both, the solid collagen construct may be in the form of fibrils. Such fibrils may be aligned due to the extrusion process such as for example, as see in FIG. 1 and FIG. 7. Such alignment can be beneficial in the preparation of implants for tissues that have similar alignment geometries. In these and other embodiments of the invention, solid collagen constructs, and processes for preparing them, are provided by extruding a suspension comprising a polymerizable collagen solution with cells, such as a polymerizable collagen oligomer solution, with an extruder to generate a solid collagen construct wherein the solid collagen construct is embedded with cells. Example of such cells include MCC, MEE, and/or ASC cells. In such embodiments, the suspension may be extruded into a container such as a die or a mold. The temperature of the extruder may be kept on ice (e.g., at about 4° C.) and the container, such as the die or mold, at room temperature or body temperature (e.g., at about 37° C.) to accelerate polymerization. The extruder may be a syringe. The polymerizable solution contains components such that polymerization can be initiated at 4° C. and accelerated at higher temperatures. For example, a typical polymerizable collagen oligomer solution may be prepared in accordance with Example 1. In many embodiments, the polymerizable collagen oligomer solution contains collagen oligomers, such as a type 1 collagen oligomer, which has been dissolved in water and acid, such as HCl. The pH is raised in the presence of one or more salts and base, such as NaOH. A sugar may optionally be added, such as glucose. An example of a combination of salts which is the one or more salts is KH_2PO_4 , $\text{Na}_2\text{HPO}_4 \cdot \text{KCl}$, and NaCl . For example, when a solubilized collagen oligomeric solution is acidic and is then neutralized by the one or more salts and the base such that the pH increases to the range of between about 4 and 10 including within about 6 and about 8, and further including about 7.4, polymerization spontaneously occurs with the rate of such polymerization being dependent upon temperature.

It may be desirable to control or identify the concentration of the collagen, such as the collagen oligomer in solution, since different concentrations may yield solid collagen constructs with different properties. Typical ranges of collagen concentrations in the polymerizable collagen solutions, such as polymerizable oligomer collagen solutions, include between about 0.1 mg/ml and about 40 mg/ml, including between about 1 mg/ml and about 10 mg/ml, including between about 2 mg/ml and about 6 mg/ml, including between about 3 mg/ml and about 5 mg/ml.

The rate of extrusion of the polymerizable collagen suspension through the extruder such as, for example, a syringe, may also be controlled. Exemplary rates include between about 1 ml/min and about 3 ml/min including, or example, about 2 ml/min. When the solid construct polymerizes, whether the suspension is in the syringe during extrusion, or after exiting the extruder such as in a container (such as a die or mold), or both, the solid collagen construct may be in the form of fibrils. Such fibrils may be aligned due to the extrusion process such as for example, as see in FIG. 1 and

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FIG. 7. Such alignment can be beneficial in the preparation of implants for tissues that have similar alignment geometries. In many embodiments, the solid collagen constructs with embedded cells may further be cultured.

The solid collagen constructs embedded with cells may be used to form tissues with aligned architectures such as muscles, nerves, tendons, or ligaments. Examples of muscle tissue include cardiac muscle, smooth muscle, skeletal muscle, and adductor muscle. Collagen constructs made with or without embedded cells may be used as implants in human or veterinary applications such as in tissues form tissues with aligned architectures such as muscles, nerves, tendons, or ligaments. Examples of muscle tissue include cardiac muscle, smooth muscle, skeletal muscle, and adductor muscle. The kits of the disclosure may contain polymerizable collagen solutions or polymerizable collagen suspension, such as from oligomeric collagen, provided that the rate of polymerization is low enough to allow for extrusion. Other kits may comprise solid collagen constructs. Such kits may be used for tissue reconstruction.

In many embodiments, the processes of the disclosure for making involves extrusion of polymerizable collagen solutions. Polymerization describes the process by which soluble collagen molecules aggregate to form solid collagen fibrils surrounded by a fluid. For example, extrusion of polymerizable collagen solutions during polymerization yields compositions where solid collagen fibrils are preferentially oriented in the direction of flow. Extrusion parameters, including temperature, oligomer concentration, flow rate, etc. can be modulated to yield different compositions. Autologous stem, progenitor, or differentiated cells and oligomer collagen for generation of different tissue compositions. For example, tissue constructs that exhibit the greatest in-vitro muscle forming activity, as measured by the extent of myoblast fusion and myotube formation, is observed with the extrusion of an appropriate high density cells cultured within a low fibril density (stiffness) matrix formed from soluble collagen oligomers. In yet another example, engineered muscle constructs can be generated by combining oligomer collagen with myoblasts cell lines, muscle progenitor cells (MPC) (for example, autologous MPCs), or motor endplate expressing MPCs (MEE). For instance, constructs fashioned from oligomers and motor endplate expressing MPCs (MEE) provide advantageous in-vivo tissue integration and muscle regeneration compared to MPC-oligomer constructs. As used herein, term “constructs” refers to solid collagen materials with or without embedded cells.

In other embodiments, solid collagen constructs such as aligned ASC-oligomer or fibrillar collagen constructs are formed by extrusion through a temperature-controlled mold, and applied to polymerizable oligomer solutions during polymerization in the presence of cells in the case of ASC-oligomer constructs or in the absence of cells in the case of fibrillar collagen constructs.

In other embodiments, 3D tissue-engineered muscle implant prepared from therapeutic cells, and type I collagen oligomers through extrusion, is provided. Examples of therapeutic cells including stem or progenitor cells (for example MPCs) or their derivatives (for example MEE MPCs). In these and other embodiments, 3D-Engineered muscle implants may be constructed from autologous cell sources and polymerizable collagen oligomers to avoid adverse immune and inflammatory responses. Such implants recapitulate the structure and functional properties of native skeletal muscle, and represents aligned muscle fibers interfacing within an appropriate, well-organized and persistent

extracellular matrix (ECM) that shows low turnover rate or high resistance to proteolytic degradation. The biological constructs integrate rapidly into host tissue in absence of immune or inflammatory response with associated neovascularization and innervation, interfacial tissue regeneration, and support scalable and patient-specific design. The ratio of cell density and the oligomer fibril density may be optimized to achieve cell-matrix physical and biochemical associations that recreate those found between muscle cells and the endomysium in vivo, resulting in accelerated in-vitro myotube formation and in-vivo muscle regeneration. For example, such solid collagen constructs made according to the disclosure can be used for reconstructing damaged muscle and cartilaginous hemilaryngeal defects as well as other muscle defects, which may result from a number of conditions including traumatic injury, tumor extraction, muscle degeneration, myopathy, and congenital malformations.

In various embodiments of the invention, the myogenic potential of MPCs may be interfaced with polymerizable collagens to create an engineered muscle for laryngeal reconstruction in the presence or in the absence of recurrent laryngeal nerve injury. For example, a MPC-oligomer construct may be extruded to achieve a fibril density that mimics those found between muscle cells and the endomysium in vivo, resulting in accelerated in-vitro myotube formation. Additional benefit of recapitulating the muscle-ECM interface was evident from the time-dependent recovery of muscle volume and function along with regeneration of supporting cartilaginous structures following implantation in vivo. Thus, for example, the advantage of the constructs of the disclosure is that they integrate with the surrounding normal tissue and immediately induce functional muscle formation within defect sites and do not trigger an inflammation response. Conventional muscle engineering strategies, which involve synthetic and natural biomaterials, are designed to degrade slowly so that they can be replaced by host deposited tissue. However, the ability of the host to generate new muscle in this situation is limited, especially since these materials typically induce inflammation.

The constructs described herein enjoy many advantages over the prior art engineered muscle tissue including i) the use of a standardized and customizable polymerizable collagen formulations, ii) the promotion of interfacial tissue regeneration by minimizing inflammation and iii) acceleration of functional muscle restoration through rapid innervation. For example, laryngeal reconstruction represents an unmet need because the methods to date are not suitable to perform such reconstructions. These prior methods have various limitations such as the lack of new muscle formation, limited vascularization and innervation which is required for functional muscle as well as triggering inflammation which delays healing. Currently, clinical strategies result in scar formation and loss of normal structure and function. By comparison, the methods and materials of the disclosure involve engineered biological muscle, prepared from solid collagen constructs embedded with cells, overcomes these limitations.

In many embodiments, and with respect to patient-specific human laryngeal muscle, the solid collagen constructs herein feature i) alignment of component MPCs and a collagen-fibril ECM rapidly upon fabrication (for example, via extrusion methods) and ii) induction of motor endplate expression to accelerate innervation following in-vivo implantation⁷. The oligomers used within the constructs are well suited for tissue engineering and regeneration strategies since they i) exhibit rapid suprafibrillar self-assembly yield-

ing highly interconnected collagen-fibril matrices resembling those found in vivo; ii) are standardized based upon their fibril-forming capacity; support cell encapsulation and distribution throughout the construct; and iv) allow customized multi-scale design across the broadest range of tissue architectures and physical properties.

SOME REPRESENTATIVE EXAMPLES

Example 1: Preparation of Collagen Composition

Type I collagen oligomers were derived from the dermis of closed herd pigs and prepared as described previously (Bailey J L, Critser P J, Whittington C, Kuske J L, Yoder M C, Voytik-Harbin S L; Collagen oligomers modulate physical and biological properties of three-dimensional self-assembled matrices, *Biopolymers* (2011) 95(2):77-93 and Kreger S T, Bell B J, Bailey J, Stites E, Kuske J, Waisner B, Voytik-Harbin S L; Polymerization and matrix physical properties as important design considerations for soluble collagen formulations, *Biopolymers* (2010) 93(8):690-707, both incorporated herein by reference). Prior to use, lyophilized collagen oligomers were solubilized in 0.01 N hydrochloric acid and rendered aseptic by chloroform exposure at 4° C. or sterile filtration using a 0.22 µm filter. A Sirius Red (Direct Red 80) assay was used to determine collagen concentration. Oligomer solutions were standardized based upon purity as well as polymerization capacity according to the ASTM international consensus standard F3089-14 (ASTM Standard F3089, 2014, "Standard Guide for Characterization and Standardization of Polymerizable Collagen-Based Products and Associated Collagen-Cell Interactions", ASTM International, West Conshohocken, PA, F3089-14, www.astm.org). Polymerization capacity is defined by matrix shear storage modulus (G') as a function of oligomer concentration of the polymerization reaction. In this way, a predictive formulary can be established that relates the concentration of a polymerizable oligomer solution to specific viscoelastic properties, namely shear storage modulus, of the resultant polymerized oligomer scaffold. Polymerization is induced using single-step neutralization with a 10x self-assembly reagent (added at a ratio of 1 part to 9 parts acidic oligomer solution) prepared according to the following recipe:

2 g KH₂PO₄ (FW 136.09)
11.5 g Na₂HPO₄ (FW 141.96)
2 g KCl (FW 74.55)
10 g glucose
80 g NaCl (FW 58.44)
20 ml 5N NaOH

It should be noted that the rate of polymerization is temperature dependent, increasing, for example over the range of 4° C. and 37° C.

Example 2: Viscoelastic Properties Testing

Viscoelastic properties of polymerized oligomer constructs were determined using oscillatory shear mode on an AR2000 rheometer (TA Instruments, New Castle, DE) as previously described (Kreger et al., 2010). Samples were polymerized on the rheometer stage for 30 min followed by a shear-strain sweep from 0.1% to 4% strain at 1 Hz. The shear storage modulus (G') at 1% strain was used as a measure of scaffold mechanical integrity.

Example 3: Myoblast Cell Line and Primary MPCs

C2C12 mouse myoblasts (ATCC, Rockville, Maryland) were cultured in Dulbecco's Modified Eagle Medium

(DMEM; Fisher Scientific, Chicago, IL) supplemented with 1% penicillin, streptomycin, amphotericin B (PSF-1; HyClone, Logan, UT), and 10% fetal bovine serum (HyClone; Logan, Utah) at 37° C. and 5% carbon dioxide. Cells were cultured to 70% confluency and used in experiments at passages 5-8. Primary MPCs were generated from skeletal muscle biopsies obtained from 12-week-old male Fischer 344 rats (Envigo, Indianapolis, IN) as previously described⁷. In brief, fresh muscle tissue was placed in myogenic growth medium (MGM), which consisted of DMEM supplemented with 1% PSF-1, 20% fetal bovine serum, and 0.1% chick embryo extract (Accurate Chemicals, Westbury, NY). Muscle was minced and digested in 0.2% collagenase type I (EMD Millipore, Temecula, CA) at 37° C. for 2 hours. Digested tissue was filtered through a 100 µm cell strainer and washed 3 times with MGM. Resulting muscle fibers were suspended in MGM, plated onto untreated 100 mm petri dishes (Fisher Scientific), and cultured overnight at 37° C. within a humidified environment of 5% carbon dioxide in air. The supernatant was removed the next morning and transferred to culture flasks (Corning Life Sciences, Corning, NY). Resultant primary muscle progenitor cells were cultured to 70% confluency and used in experiments at passages 3 to 5.

Example 4: Conventional Casting of Tissue Constructs

Tissue constructs were prepared by conventional casting methods resulting in composites comprising randomly organized cells and collagen-fibril scaffolds. Rat primary MSCs prepared according to Example 3 were suspended at a density of 106 cells/mL in neutralized oligomer solutions (1.5 mg/mL). Neutralization was achieved using multi-step or single-step procedures and reagents as described in Example 1. Neutralized MSC-oligomer suspensions were maintained at 4° C. prior to use. The MSC-oligomer suspension was aliquoted into a 24-well plate (500 µL/well) and subsequently polymerized at 37° C. for 10 minutes. Once polymerized, constructs were cultured for 24 hours in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 1% penicillin, streptomycin, amphotericin B (PSF-1; HyClone, Logan, UT) and 10% fetal bovine serum (HyClone; Logan, Utah) at 37° C. within a humidified environment of 5% carbon dioxide in air. Tissue constructs were fixed in 3% paraformaldehyde of culture and stained with phalloidin for visualization of the actin cytoskeleton. For 3D qualitative analysis, tissue constructs were imaged using an Olympus FluoView FV-1000 confocal system adapted to an inverted microscope (IX81, Olympus Corporation, Tokyo, Japan). Confocal image stacks were processed using Imaris software and images analyzed on ImageJ for alignment using the Directionality algorithm. As shown in FIG. 3D-F, tissue constructs formed by conventional casting methods exhibited randomly organized cells and collagen fibril scaffolds.

Example 5: Preparation of Aligned Muscle Constructs by Extrusion

Acidic oligomer solutions were prepared and standardized as described in Example 1. Neutralized oligomer solutions were prepared according to Example 2 at a final concentration of 1.5 mg/mL and maintained at 4° C. to slow polymerization rate. MPCs were then suspended in the neutralized oligomer solution at 10⁷ cells/mL and maintained at 4° C. prior to use. The 4° C. temperature was

maintained by placing the solutions and suspensions on ice. The MPC-oligomer suspension (500 µL) was drawn up into a syringe and then extruded (FIG. 1) from the syringe (3 cc) at a rate of 2 mL/min into an 4-mm diameter cylinder mold prepared from Ultem (FIG. 2A and open on both ends), which was maintained at 37° C. and stayed in the mold for about 10 minutes to form "constructs". Afterwards, the constructs were transferred to a customized culture chamber and secured in tension with fasteners (FIG. 4A) and cultured in DMEM supplemented with 10% fetal bovine serum and 1% PSF-1 at 37° C. in 5% carbon dioxide in air. After 24 hours, constructs were stained with phalloidin (actin) and Draq5 (nuclei) and analyzed by confocal microscopy to visualize solid collagen-fibril scaffold and cells within the constructs. Confocal image stacks were analyzed on ImageJ for alignment using the Directionality algorithm. As indicated in FIG. 3, solid collagen fibrils (FIG. 3A) and cells (FIG. 3B) were highly aligned parallel (FIG. 3C) to the direction of flow. By contrast, constructs prepared by conventional casting methods (Example 4) exhibited random organization (FIG. 3F) of solid collagen fibrils (FIG. 3D) and cells (FIG. 3E).

Example 6: Effect of Cell Density and Oligomer Concentration on Engineered Muscle Constructs Prepared by Extrusion

The same extrusion process described in Example 5 was applied for C2C12 myoblasts at densities of 10⁶ and 10⁷ cells/mL as set forth in this Example 6. C2C12 myoblasts were suspended at each density in neutralized oligomer solutions (Example 1) prepared at either 1.3 mg/mL or 3.3 mg/mL. Following extrusion, constructs were cultured within the custom culture device for 14 days under passive tension (FIG. 4A) prior to fixation and staining with phalloidin and Draq5 for visualization of cellular actin and nuclei, respectively. As indicated in FIG. 4, constructs prepared at high cell density and low oligomer concentration (Panel c of FIG. 4B) exhibited the necessary cell-cell interactions and collagen-fibril scaffold support to facilitate a high level of cell fusion, nuclear alignment, and myotube formation. By contrast, constructs prepared at low cell density and high oligomer concentration (Panel b of FIG. 4B) exhibited the least amount of cell fusion and alignment largely owing to the increased solution viscosity and density of solid fibrils formed during polymerization. Follow-up verification studies revealed that this fabrication strategy was also applicable to primary rat MPCs, yielding highly reproducible engineered muscle with viable cells distributed throughout the construct after 2 weeks of culture (FIG. 5).

Example 7: Generation of Aligned Muscle Construct with Motor Endplate Expressing MPCs (MEE)

The same extrusion process outlined in Example 6 was applied to create aligned muscle constructs where the MPCs were induced to express motor endplates. Briefly, following extrusion constructs were cultured for 5 days after which time motor endplate expression was induced by adding acetylcholine chloride (40 nM; Tocris Bioscience, Bristol, England), agrin (10 nM; R&D Systems, Minneapolis, Minnesota), and neuregulin (2 nM; R&D Systems) to the culture medium. Constructs were cultured an additional 7 days with medium changes every 3 days. Motor endplate expression

was confirmed by immunostaining with Alexa Fluor 594 conjugated α -bungarotoxin (Molecular Probes, Eugene, Oregon).

Example 8: Aligned Tissue Constructs Prepared by Extrusion of Human Adipose-Derived Stem Cells (ASCs) Suspended in Polymerizable Oligomer Solutions

The same extrusion process described in Example 5 was applied to human adipose-derived stem cells (ASCs). ASCs represent an attractive autologous cell type for tissue engineering and regenerative medicine applications since they can be easily harvested from subcutaneous fat via conventional liposuction techniques. ASCs represent a mesenchymal stem cell source with self-renewal property and multipotential differentiation. Said differently, these cells are exhibit a high proliferative capacity and the ability to differentiate into multiple different cell types, including adipocytes, osteoblasts, tenocytes, myocytes, and neurocytes. Low passage human ASCs (Lonza) were cultured within collagen oligomer-coated flasks and maintained in complete medium comprising EGM-2 supplemented with 12% Hyclone FBS. Prior to extrusion, ASCs were suspended at a density of 500,000 cell/mL in neutralized oligomer solution (1.5 mg/mL). For comparison purposes, randomly organized oligomer-ASC constructs were prepared within standard 96-well plate using conventional casting methods as described in Example 4. Following culture for 24 hours, constructs were fixed in paraformaldehyde, stained with phalloidin (F-actin) followed by Draq5 (nuclear) for 30 minutes at room temperature, and imaged using confocal microscopy. Confocal image stacks were processed using Imaris software and images analyzed on ImageJ for alignment using the Directionality algorithm. As shown in FIG. 6A-B, tissue constructs prepared by extrusion showed uniform alignment of cells in parallel with the direction of flow. By contrast, tissue constructs prepared by conventional casting techniques exhibited randomly organized cells with no preferred directionality (FIG. 6C-D). Such randomly organized cells fail to recapitulate the structure and functional properties of native skeletal muscle.

Example 9: Creation of Aligned Collagen-Fibril Construct Via Extrusion

The same extrusion process described in Example 5 was applied to oligomer solutions in absence of cells as set forth in this Example 9. Stock oligomer solution was diluted with 0.01N hydrochloric acid and neutralized as described in Example 1 to achieve a neutralized oligomer solution at a concentration of 3.3 mg/ml. The neutralized oligomer solution was kept on ice (4° C.) prior to induction of polymerization. Oligomer solution (500 μ L) was extruded from a syringe (3 cc) at a rate of 2.1 mL/min into an Utem 4-mm diameter cylinder mold, which was maintained at 37° C. Following extrusion, the construct was imaged via confocal reflection microscopy for visualization of the solid fibril architecture. As shown in FIG. 7, the resultant fibrillar collagen scaffold was highly aligned.

Example 10: Laryngectomy and Implantation of Engineered Muscle

Engineered muscle constructs formed by extrusion of primary rat MPCs suspended within neutralized oligomer solutions were evaluated in an established rat partial laryn-

gectomy model. Neutralized oligomer solutions were prepared as described in Example 1 at a final concentration of 1.3 mg/mL. Primary rat MPCs were generated according to Example 3 and suspended in the neutralized oligomer solution at a density of 10E7 cells/mL. The suspension was extruded to form MPC-oligomer constructs as described in Example 6. Constructs were cultured for 5 days with medium changes every 2 days. On day 5, medium was changed to differentiation medium, representing DMEM supplemented with 8% horse serum (HyClone) and 1% PSF-1. Constructs were cultured for an additional 7 days to induce myotube formation. MEE-oligomer constructs were extruded as described above and cultured as described in Example 7. Oligomer only constructs were prepared by extrusion of neutralized oligomer solutions (1.3 mg/mL) in absence of cells.

The animal study protocol was approved by Purdue Animal Care and Use Committee, and institutional guidelines, in accordance with the National Institutes of Health guidelines, were followed for the handling and care of the animals. 12 Fisher 344 rats were anesthetized with intraperitoneal injection of xylazine and ketamine and then maintained on 1-4% isoflurane. The ventral larynx was exposed via a midline incision. The sternohyoid muscle was incised and reflected to expose the thyroid cartilage. A section (approximately 2 mm $\times$ 2 mm) of thyroid cartilage and associated adductor muscle was removed from the left side. Animals were randomized into the following experimental groups: MPC-oligomer construct (n=4), MEE-oligomer construct (n=4), oligomer construct only (n=2), and defect only control (n=2). The sternohyoid muscle was reapposed and sutured, and subcutaneous tissue and skin closed with 5-0 Vicryl suture.

Example 11: Laryngectomy in the Presence and Absence of Recurrent Laryngeal Nerve Injury and Implantation of Engineered Muscle and Cartilage

Engineered cartilage and muscle constructs were implanted in an established rat partial laryngectomy model in the presence and absence of recurrent laryngeal nerve injury. Engineered muscle constructs were formed by extrusion of primary rat MPCs suspended within neutralized oligomer solutions. Neutralized oligomer solutions were prepared as described in Example 1 at a final concentration of 1.3 mg/mL. Primary rat MPCs were generated according to Example 3 and suspended in the neutralized oligomer solution at a density of 10E7 cells/mL. The suspension was extruded to form MPC-oligomer constructs as described in Example 6. Constructs were cultured for 5 days with medium changes every 2 days. On day 5, medium was changed to differentiation medium, representing DMEM supplemented with 8% horse serum (HyClone) and 1% PSF-1. Constructs were cultured for an additional 7 days to induce myotube formation. MEE-oligomer constructs were extruded as described above and cultured as described in Example 7.

The animal study protocol was approved by Purdue Animal Care and Use Committee, and institutional guidelines, in accordance with the National Institutes of Health, were followed for the handling and care of the animals. Fisher 344 rats were anesthetized with intraperitoneal injection of xylazine and ketamine and then maintained on 1-4% isoflurane. The ventral larynx was exposed via a midline incision. The sternohyoid muscle was incised and reflected to expose the thyroid cartilage. A section (approximately 2 mm $\times$ 2 mm) of thyroid cartilage and associated adductor

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muscle was removed from the left side. Animals were randomized into groups receiving engineered constructs with MPC or MEE cells with and without RLN injury. All groups received identical engineered cartilage implants with endpoints at 1, 3, and 6 months. The muscle construct was placed into the defect (similar to a medialization laryngoplasty implant), followed by the cartilage construct over top with extrusion prevented by suturing overlying sternohyoid muscles over the cartilaginous defect. For groups with RLN injury, the left recurrent laryngeal nerve was cauterized as it entered the larynx. The subcutaneous tissue and skin were then closed with 5-0 Vicryl suture.

Example 12: Video Laryngoscopy and Laryngeal Electromyography (EMG)

Video laryngoscopy and/or laryngeal electromyography were performed on anesthetized rats at 1, 3, and 6 month timepoints following partial laryngectomy and reconstruction as described in Examples 10 and 11. Video laryngoscopy was performed using a rigid endoscope with attached camera. Electromyogram (Niking Viking Quest electromyography machine, Madison, Wisconsin) was used with a 25-gauge bipolar concentric needle, settings with amplitude of 50 to 100 μ V, 10- to 100-ms sweep speeds, and a grounding clamp at the exposed lateral sternocleidomastoid muscle. The EMG recording needle was inserted directly into the center of the defect/implant site, the adductor (TA) muscle complex, and the posterior cricoarytenoid (PCA) muscle during laryngospasm and at rest. Immediately following laryngoscopy and electromyography procedures, rats were humanely euthanized and tissue collected.

Example 13: Histopathological and Histochemistry Assessment

Histopathological analysis was performed at 1, 3, or 6 month timepoints following partial laryngectomy and reconstruction as described in Examples 10 and 11. After euthanasia, rat larynges and associated implants were harvested en bloc, fixed in 4% paraformaldehyde overnight, and then transferred to 30% sucrose at 4° C. for an additional 24 hours. Cryosections (25 μ m thickness) were prepared on a Thermo Cryotome FE (Fisher Scientific, Kalamazoo, MI). Sections were stained with hematoxylin & eosin (H&E) and Alcian blue for histopathological analysis. Slides were viewed on a Nikon upright microscope (Eclipse E200, Nikon, Melville, NY) and images captured with a Leica camera (DFC480, Leica Buffalo Grove, IL). Myofiber diameter was measured using Image J software (NIH).

For histochemistry analysis, all specimens were washed with phosphate buffered saline (PBS) 3 times, permeabilized with 0.1% Triton X-100 for 20 minutes, and blocked with 1% BSA for 2 hours. Whole engineered muscle constructs were incubated overnight at 4° C. with Alexafluor 488 conjugated phalloidin (1:25 Molecular Probes, Eugene, OR) for visualization of F-actin and counterstained with Draq5 (1:1000 Cell Signaling, Danvers, MA 4084L) nuclear stain. For staining tissue explant cryosections, beta III tubulin conjugated primary antibody (1:10 NL647 Molecular Probes) was applied and incubated overnight at 4° C. After rinsing extensively, slides were incubated with Alexa Fluor 594 conjugated α -bungarotoxin (1:100) for 2 hours at room temperature. Slides were rinsed and mounted with Fluorogel for imaging on an Olympus Fluoview confocal microscope

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(IX81, Olympus Waltham, MA) or Zeiss LSM 880 confocal microscope (Oberkochen, Germany).

Example 14: Reconstruction of Laryngeal Defects with MPC-Oligomer Constructs Yields Myogenesis, Neurovascular Regeneration, and Tissue Integration in Absence of an Inflammatory-Mediated Foreign Body Response

Laryngectomy and implantation of engineered muscle was performed as described in Example 10. During the post-surgical period, all animals steadily gained weight and showed no signs of laryngeal compromise. As expected, partial laryngectomy with resection of cartilage and muscle and no treatment resulted in a healing response marked by inflammation and fibrous tissue formation (scar tissue) within the defect area (FIG. 8B). In contrast, oligomer implants, with and without MPCs, integrated rapidly with adjacent host tissues, showing no significant inflammatory reaction or proteolytic degradation (FIG. 8B-E). A surprising finding was that chondrogenesis accompanied muscle regeneration over time (FIG. 8E). Compared to the oligomer only group, MPC-oligomer and MEE-oligomer constructs showed more rapid muscle regeneration and maturation as evidenced by obvious striations within the implants as early as 1 month that became more evident at 3 months. By 3 months, the relative extent of striated muscle increased, and was supported by neurogenesis, as evidenced by prominent beta III tubulin staining arising from the graft-host tissue interface (FIG. 10). Interestingly, results based on beta III tubulin and α -bungarotoxin staining indicated that the greatest level of innervation was achieved with the MEE-oligomer group. Qualitative EMG measurements provided further corroborating evidence of enhanced innervation of the MEE-oligomer constructs, with the MEE-oligomer group demonstrating the greatest level of motor unit activity (recruitment) during laryngospasm (FIG. 9).

Results of these studies suggest that aligned MEE-oligomer constructs contributed to rapid tissue integration and regeneration in the absence of any significant inflammatory reaction or rapid proteolytic degradation, yielding restored muscle with enhanced innervation on histology, and functional elicitation of motor unit potentials. Major advantages of the model include the use of autologous cells, aligned muscle constructs for rapid restoration of vascularization, innervation, and function, and a scalable fabrication method that can be translated into patient-specific designs.

Example 15: Reconstruction of Denervated Hemilarynx after Partial Laryngectomy with Engineered Cartilage and Muscle Constructs Yields Myogenesis, Neurovascular Regeneration, and Functional Tissue Restoration in Absence of an Inflammatory-Mediated Foreign Body Response

Laryngectomy in the presence and absence of recurrent laryngeal nerve injury and implantation of engineered muscle and cartilage as described in Example 11. All animals survived the post-surgical period with no life-threatening complications. Some RLN injured animals showed mild stridor but this resolved with time. All animals steadily gained weight over the study period. Post-mortem gross pathological exam showed integration of the cartilage and muscle implants into host tissue with no macroscopic signs of inflammation. Alcian blue staining of cryo-sectioned specimens was weakly positive at 1 month (FIG. 11B). At 3 and 6 month timepoints, sections demonstrated progressive

increase in Alcian blue staining consistent with glycosaminoglycan deposition and cartilage formation (FIG. 11C-D). H&E staining at 1 month displayed immature muscle neighboring native adductor muscle with no foreign body response (FIG. 12A,D). At 3 months, maturation of the muscle within the implant was demonstrated by development of cross striations and myofiber alignment (FIG. 12B, E). By 6 months, little to no immature muscle was visible within the defect area suggesting complete integration with nature tissue (FIG. 12C-F).

α -bungarotoxin and beta III tubulin staining demonstrated that implants that were induced to form motor endplates (MEE) showed a great number of motor endplates and robust innervation (FIG. 13). Qualitative EMG measurements of the adductor muscle complex of the MPC-oligomer treated animals showed low level activity (FIG. 14A,B) whereas the MEE-oligomer treated animals showed near normal muscle activity, with interference and recruitment patterns (during laryngospasm) that mimicked normal adductor muscle (FIG. 14C,D). This finding was further corroborated by adductor muscle complex myofiber diameter measurements, with values for MEE-oligomer treated animals significantly greater than those for MPC-oligomer treated animals ($p < 0.001$ FIG. 14E).

In nerve injured animals, video laryngoscopy showed function recovery in 100% of animals receiving the MEE implant with definitive although slightly delayed movement at all timepoints. By contrast, none of the animals receiving the MPC implant showed definitive movement at any timepoint (FIG. 15).

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We claim:

1. A method of enhancing the repair of damaged tissues selected from muscle, tendon, ligament, and nerve, said method comprising the implantation of a fibrillar collagen

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network having a uniform density of aligned polymerized oligomeric collagen fibrils throughout said network, at the site in need of tissue repair.

2. The method of claim 1 wherein said implantation comprises the steps of extruding a polymerizable oligomeric collagen solution at or adjacent to said site in need of tissue repair, wherein said polymerizable oligomeric collagen solution is extruded under conditions permissive to polymerization of the oligomeric collagen to form said fibrillar collagen network having a uniform density of polymerized oligomeric collagen fibrils throughout said network.

3. The method of claim 2 wherein the polymerizable oligomeric collagen solution is extruded via the use of a syringe as the extruder.

4. The method of claim 3 wherein the polymerizable oligomeric collagen solution is polymerized during the extrusion to form fibrils, wherein said fibrils are aligned during extrusion.

5. The method of claim 3 wherein the polymerizable collagen solution is extruded at a rate of between about 1 ml/minute and about 3 ml/minute.

6. The method of claim 1 wherein said fibrillar collagen network comprises oligomeric collagen and monomeric collagen, wherein said monomeric collagen comprises one or both telocollagen molecules and atelocollagen molecules.

7. The method of claim 1 wherein said fibrillar collagen network further comprises cells.

8. A method of accelerating muscle restoration or promoting interfacial tissue regeneration, said method comprising the implantation of a fibrillar collagen network having a uniform density of aligned polymerized oligomeric collagen fibrils throughout said network, at the site in need of muscle restoration or tissue regeneration.

9. The method of claim 8 wherein said implantation comprises the steps of extruding a polymerizable oligomeric collagen solution at or adjacent to said site, wherein said polymerizable oligomeric collagen solution is extruded under conditions permissive to polymerization of the oligomeric collagen to form said fibrillar collagen network having a uniform density of polymerized oligomeric collagen fibrils throughout said network.

10. The method of claim 8 wherein said fibrillar collagen network comprises oligomeric collagen and monomeric

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collagen, wherein said monomeric collagen comprises one or both telocollagen molecules and atelocollagen molecules.

11. The method of claim 8 wherein said fibrillar collagen network further comprises cells.

12. The method of claim 11, wherein the cells are embedded within the fibrillar collagen network.

13. The method of claim 12, wherein the cells are stem cells, progenitor cells, motor endplate-expressing muscle progenitor cells, or a combination thereof.

14. A method of enhancing the repair of damaged tissues, said method comprising

injecting a polymerizable oligomeric collagen solution into, or adjacent to, a site in need of tissue repair, said injection step comprising providing an extruder that contains said polymerizable oligomeric collagen solution and extruding said polymerizable oligomeric collagen solution through said extruder and into, or adjacent to, the site in need of tissue repair under conditions permissive to polymerization of the oligomeric collagen to form a fibrillar collagen network having a uniform density of polymerized oligomeric collagen fibrils throughout said network at, or adjacent to, said site in need of tissue repair.

15. The method of claim 14 wherein the extruder is a syringe.

16. The method of claim 15 wherein the polymerizable oligomeric collagen solution is polymerized during the extrusion to form a fibrillar collagen network having a uniform density of aligned polymerized oligomeric collagen fibrils throughout said network.

17. The method of claim 16 wherein the polymerizable oligomeric collagen solution is extruded at a rate of between about 1 ml/minute and about 3 ml/minute.

18. The method of claim 14 wherein said fibrillar collagen network comprises oligomeric collagen and monomeric collagen, wherein said monomeric collagen comprises one or both telocollagen molecules and atelocollagen molecules.

19. The method of claim 14 wherein said polymerizable oligomeric collagen solution further comprises cells.

20. The method of claim 19, wherein the cells become embedded within the formed fibrillar collagen network.

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